Supplementary Information for
Polymer-inspired mechanical metamaterials

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Supplementary Note 1. The polymeric meta-structure generation algorithm

We developed an object-oriented algorithm within a Python environment, leveraging the RhinoScriptsyntax library in Rhino 6, for the construction of polymeric meta-structures. The schematic representation of the polymeric meta-structure programming is depicted in Supplementary Fig. 22(a). Drawing parallels to a polymer's chemical structure, we designed a complex virtual polymer object characterized by principal attributes—namely, polymer chains and crosslinks that define its geometric configuration—and secondary attributes that govern its programmable properties, such as boundary conditions, molecular and crosslink densities, and proto-crystalline levels. These principal attributes are encapsulated within child objects, inheriting the parent polymer's characteristics. The geometry of each polymer chain is determined by its length and variability, with its contours derived from the parent polymer's boundaries. Each polymer chain comprises structural nodes that delineate its spatial trajectory. Crosslink objects are formed at the juncture of two parent nodes belonging to distinct chains. Supplementary Fig. 22(b) in the Supplementary Information outlines the algorithm's workflow. The parameters for the number of crosslinks ($N_{crosslinks}$), chain length (l), and number of chains (N_{chains}) are derived from the following set of equations:

$$N_{crosslinks} = n_c V_{boundary} \tag{S1}$$

$$l = l_{load}/p \tag{S2}$$

$$N_{chains} = \rho V_{boundary} \tag{S3}$$

where $V_{boundary}$ is the volume within the boundary of polymer, n_c is the crosslink density, ρ is the molecular-like density, p represents the proto-crystalline level, l_{load} is the projection length of a polymer chain along the load direction. The edge nodes of each polymer chain are randomly instantiated within the confines of the polymer's boundary, adhering to specific constraints to guarantee the attainability of the prescribed polymer chain length. These constraints are meticulously designed to maintain the structural integrity and feasibility of the polymer chain configuration within the defined spatial parameters:

$$D(p_{start}, p_{end})S \leq l \tag{S4}$$

where $D(p_{start}, p_{end})$ is the spatial distance between the start (p_{start}) and end edge nodes (p_{end}), S is the safety factor considering the potential extra lengths required for a chain to bypass existing chains. Parameters such as l, N_{chains}, and sets of (p_{start}, p_{end}) are input into the polymer chain generation algorithm to synthesize the requisite number of intricate, helical chains. Concurrently, the number of crosslinks ($N_{crosslinks}$) and the synthesized chains are fed into the crosslinking algorithm to form the polymer's crosslinks. These components are then unified through a Boolean operation to yield the final polymeric meta-structure.

The procedural intricacies and methodologies underlying the polymer chain generation are elucidated in Supplementary Fig. 23(a-b) for the algorithmic framework and Supplementary Fig. 23(c) for the workflow. The chain's variability is governed by a fluctuation factor ($N_{fluctuation}$), which dictates the count of control nodes responsible for imparting torsion to the chain. The maximal permissible fluctuation is defined in accordance with the thresholds established through an additive-manufacturing-oriented manufacturability analysis, as delineated in our preceding work [1]. This constraint enables the emulation of the high degrees of torsion and stochasticity characteristic of natural

polymeric molecular configurations. In light of the maximal fluctuation, the chain's length, and the spatial disposition of the terminal nodes, we can derive the fluctuation step ($S_{fluctuation}$) and the node region size S_{node}). These calculations are instrumental in ensuring the fidelity of the chain's fluctuation and its conformance to the specified length, as outlined by the following equations:

$$S_{fluctuation} = D(p_{start}, p_{end})/N_{fluctuation} \quad \text{(S5)}$$

$$S_{node} = 2\sqrt{(l/N_{fluctuation})^2 - S_{fluctuation}^2} \quad \text{(S6)}$$

The polymer chain will be generated from a number of node regions climbing from the start node to the end node separated by the fluctuation step. To ensure the reachability of the end node, the nodes should be generated in a feasible chain region described by two cones with the height ($H_{feasible}$) and radius ($R_{feasible}$) specified by:

The generation of the polymer chain proceeds through a sequence of node regions extending from the start node to the end node, each spaced apart by the fluctuation step. To secure the attainability of the end node, nodes are established within a viable chain region demarcated by two cones. The dimensions of these cones are provided in equations (S7-S8), specifically their height ($H_{feasible}$) and radius ($R_{feasible}$), are delineated to ensure that the chain's trajectory remains within a constructible domain, facilitating the seamless connection from the commencement to the termination of the chain. This approach guarantees that the polymer chain adheres to the spatial constraints imposed by the geometrical configuration and the inherent flexibility of the polymer structure, thereby ensuring the integrity and functionality of the resulting polymeric meta-structure.

$$H_{feasible} = D(p_{start}, p_{end})/2 \quad \text{(S7)}$$

$$R_{feasible} = S_{node}N_{fluctuation}/4 \quad \text{(S8)}$$

Each node within a polymer chain is instantiated within a nodal region defined by a circular area centered on the preceding node, elevated by a height equivalent to the fluctuation step. The viable nodal region is identified as the intersection between the feasible chain region and this nodal region, with an allowance for exclusion zones (safe regions) around existing chains to preclude inadvertent inter-chain collisions. The radius of these safe regions is set to twice the radius of the chain itself, ensuring a buffer that maintains the structural integrity and spatial separation of adjacent chains. This methodology ensures that each chain node not only contributes to the overall structural complexity of the polymer but also respects the spatial constraints imposed by the proximity of neighboring chains, thereby optimizing the polymeric meta-structure's functional and aesthetic attributes. The precise relationships governing these spatial considerations are articulated in the subsequent formulation:

$$P_{node,feasible} = \{\forall p | p \in P_{node} \cap P_{chain,feaisble}, p \notin P_{existing\ chains}\} \quad \text{(S9)}$$

where $P_{node,feasible}$ is the sets of feasible node positions, p represents the node positions, P_{node}, $P_{chain,feaisble}$, and $P_{existing\ chains}$ is the sets of node positions of the node region, feasible chain region, and safe region of existing chains, respectively. Upon the generation of a new node, the polymer chain is updated by designating this node as the current one. The continuity and smoothness of the chain geometry are ensured by employing the AddInterpCurve function from the rhinoscriptsyntax, facilitating the creation of seamless transitions between nodes. When the terminal node is reached, the chain is deemed complete and is subsequently incorporated into the collection of existing chains. This iterative process continues until the predetermined quantity of chains is achieved, thereby

fulfilling the required molecular-like density and concluding the assembly of the polymer chains.

The design and procedural framework of the crosslinking algorithm are depicted in Supplementary Fig. 24(a) and Supplementary Fig. 24 (b). Within the assemblage of generated polymer chains, the nodes from each chain are categorized and color-coded for distinction in the schematic representation. A node is randomly selected from these categories to serve as the starting point for a crosslink. To prevent intra-chain crosslinking, the terminal node for each crosslink is chosen from a different category, ensuring it is the nearest neighbor to the start node but not on the same chain. This method of selection and linkage results in the formation of a new crosslink, which is then added to the collective set of crosslinks. To avoid redundancy, the starting node of each newly formed crosslink is removed from its category, precluding the generation of duplicate crosslinks. This procedure is repeated until the target number of crosslinks is reached, ensuring the desired crosslink density is attained, thereby completing the structural framework of the polymeric meta-structure.

Supplementary Note 2. Design and manufacturing of comparison lattice structures

A series of representative lattice architectures were designed to enable a consistent comparison of mechanical performance across different topological classes, including plate-based lattices, shell-based triply periodic minimal surface (TPMS) structures, two-dimensional directional lattices, and crystal-inspired metamaterials (CIMs). All structures were constructed with geometric parameters such that their relative densities fall within a comparable range. Plate-based lattices, including simple cubic (SC), body-centered cubic (BCC), and face-centered cubic (FCC) topologies, were generated with a unit cell size of 5 mm, where the plate thickness was varied from 0.2 to 0.5 mm to ensure thin-plate behavior while maintaining comparable densities. TPMS lattices with a gyroid topology were constructed using a unit cell size of 5 mm, with shell thickness varied from 0.2 to 0.8 mm to span a density range comparable to that of the plate lattices while preserving thin-shell characteristics. Three representative two-dimensional lattice topologies, including honeycomb, re-entrant honeycomb, and triangular lattices, were generated with a characteristic cell size of 5 mm, and the strut thickness was varied from 0.2 to 0.8 mm to achieve comparable densities. CIMs were designed based on topologies including tesseract, BCC, FCC, octet-truss, and vintiles architectures, with a unit cell size of 5 mm. The strut diameter was varied from 0.6 to 1.2 mm to ensure that the resulting relative densities covering those of the proposed polymer-inspired metamaterial (PIM) architectures, enabling a fair comparison across different structural classes. Twisted variants of the CIMs were constructed by introducing helical substructures into the lattice framework. Two configurations were constructed on BCC, FCC, and octet-truss frameworks, including pure woven structures consisting solely of twisted filaments and composite structures in which the twisted elements were integrated with the underlying straight crystal-inspired lattice beams. To accommodate the geometric constraints associated with helical deployment and to avoid unintended contact during fabrication, a unit cell size of 10 mm was adopted with strut diameter at 0.5 mm. Each twisted unit contains three helical filaments with a pitch of 1 mm and a radial distance of 2 mm from the central axis, ensuring large twisting without structural interference or fusion during the printing process. Compared with the triangular, re-entrant honeycomb, and honeycomb lattices shown in Supplementary Fig. 4, the strengthened PIMs achieve enhanced directional performance through the coupled strengthening effects of pre-stretch chain alignment, molecular-density strengthening, crosslink strengthening, and crystalline/proto-crystalline ordering. The designed structures are fabricated, post-processed, and tested in same configurations of PIMs to ensure fair comparison.

Supplementary Note 3. The design of programmable humanoid tissue and its robotic components.

The design of humanoid tissues facilitated by PIM mirrors the complexity of human tissues, as depicted in Supplementary Fig. 16(a). These programmable humanoid tissues incorporate a tendon-like structure that anchors to the robot's rigid components, alongside a muscle-like region that offers softer, programmable connections. This design emulates the dynamic stress responses of human muscles, which exhibit varying reactions under different strain conditions, whether in a state of relaxation or contraction.

To mimic the nuanced mechanical behavior of human tissues, the humanoid tissues are engineered with a molecular chain strain factor (f_c) ranging from 20% to 50%. This variability allows for the simulation of complex movements, such as walking and finger bending, illustrated in Supplementary Fig. 16(b-c). For instance, the upper region of a robotic toe is infused with tissues programmed at f_c = 20%, ensuring the replication of human-like toe flexion. Additionally, a gradient of f_c values defined as df_c/dx, spanning from 0 mm^{-1} to 1.87 mm^{-1}, is strategically implemented along the toe to intricately program the walking mechanics. Here, x represent the distance vertically to the toe PIM connection direction. In parallel, the design of the humanoid finger tissues incorporates varied f_c values across different regions—namely, the side, back, front, front-side, and back-side—of the finger connections. This distribution facilitates the engineering of finger bending mechanics that closely emulate human finger movements, accommodating various directions of motion. This sophisticated design strategy ensures that the robotic appendages not only mimic the appearance of human limbs but also closely replicate their functional mechanics and responsiveness.

Supplementary Note 4. The design of CIM+PIM composite and its robotic components.

The design blueprints and the composite of CIM+PIM are illustrated in Supplementary Fig. 12(a). In this research, the octet-truss lattice configuration is adopted as the crystal meta-structure due to its exceptional stiffness characteristics. It is imperative to acknowledge, however, that the methodology proposed herein is adaptable to various topologies, not exclusively the octet-truss structure. On the external surfaces of the crystal meta-structures, random attachment points are designated for linkage with the polymer chains. The quantity of these attachment points, denoted as $N_{attach\ points}$, positioned on specific exterior surfaces, is calculated based on the polymer's molecular-like density and the volume of the flexible framework ($V_{framework}$) adjacent to that surface. This correlation is quantitatively expressed in equation (S3), establishing a direct relationship between the molecular-like density, the framework's volumetric parameters, and the distribution of attachment points, thereby ensuring an optimal integration of the polymer chains with the crystal meta-structure.

$$N_{attach\ points} = \rho V_{flexible} \tag{S10}$$

Crystal meta-structures are strategically positioned at locations of critical stress within the design to enhance structural rigidity where it is most needed, seamlessly integrating with the surrounding flexible framework. This flexible framework is intricately woven with polymer chain networks, employing the PSG algorithm to finely tune the crosslink density. This careful adjustment allows for a tailored balance between mechanical flexibility and stiffness, ensuring the structure meets specific performance criteria.

These meta-structures are embedded within the overarching conglobate structure, offering robust compressive protection against potential damage. Flexible regions, designed to facilitate complex movements akin to an exoskeleton's articulations, maintain their ability to alter mechanical postures while being infused with polymer chains, as demonstrated in Supplementary Fig. 12(b). This design principle extends to specialized applications such as the multimodal grasping finger depicted in Supplementary Fig. 12(c), where crystal meta-structures at the attachment points augment grasping efficacy. Concurrently, the surrounding polymer frameworks afford the finger versatility in geometric adaptation, enabling it to conform to a variety of objects. This intricate interplay of rigid and flexible components exemplifies a sophisticated approach to designing structures that combine high strength with adaptable form and function.

Supplementary Note 5. Design programmability of PIM along different material orientations.

The macro-chains and cross-links of the polymer are highly engineerable, enabling the design of programmable mechanical behaviors tailored to a wide range of functional applications. Here, two case studies are presented to demonstrate property programmability along multiple material orientations and the realization of unique functional responses.

Supplementary Fig. 25a shows the microstructure of a PIM programmed with a p = 57% along the −22.5° and 22.5° material orientations. This directional programmability is achieved by artificially engineering the macro-chains such that their shared lengths in the specified directions correspond to the targeted proto-crystalline level. Experimental results (Supplementary Fig. 25b) confirm the bi-directional mechanical performance of the PIM, with moduli of 0.65 and 0.64 MPa, respectively. Supplementary Fig. 25c also shows that the samples can be programmably designed to achieve desired modulus and strength along both the printing direction and the direction perpendicular to it. This level of programmable mechanics control not only ensures multi-directional mechanical stability under practical engineering conditions but also offers opportunities to design polymer-based metamaterials with orientation-specific property distributions for application-specific requirements.

In addition, we demonstrate functional mechanics programmability through the engineering of crack propagation paths, highlighting the potential to mitigate long-standing challenges in the fracture control of artificial materials. Supplementary Fig. 6a illustrates the distribution of cross-links designed to guide damage along a semi-circular crack path by locally varying material strength. DIC measurements were conducted to capture strain distributions during fracture testing at a displacement rate of 2 mm/min. The experimental outcomes (Supplementary Figs. 6b-c) validate the effectiveness of the approach, showing that fracture strains were successfully confined along the intended paths, with fracture strain measured to be 43x higher in the crack path region compared to the surrounding material.

Supplementary Note 6. Macroscale energy partition modelling of PIMs

To interpret the polymer-inspired strengthening with macroscale architectural mechanics, and systematically program the PIM mechanics using programmable design algorithm, an energy partition framework is established to characterize the resulting mechanical responses. Each architected polymer-inspired chain is represented as a slender fiber of contour length L, with centerline coordinate $s \in [0, L]$ and circular cross-section of area A. The material of the fiber has axial modulus $E_{\mathrm{mat}}$and shear modulus G_{mat}. Its resistance to bending and twisting is characterized by a second moment of area for flexion $I_{\mathrm{flex}}$and a polar moment of inertia J. The kinematics of the polymer-inspired chain are described by three fields along the contour. The scalar $u(s)$denotes the displacement of the material along the local tangent direction, and encodes tensile extension. The rotation $\theta_{flex}(s)$describes how the centerline bends and unbends, through its derivative $\kappa(s) = \partial\theta_{flex}/\,\partial s$, which is the local curvature. The twist angle $\theta_{reorient}(s)$describes the rotation of the cross section about the chain axis, which is the kinematic mechanism behind torsional reorientation of the segments and local rotation at entanglement-like nodes. The energy densities associated with these three modes are

$$w_{debond}(s) = \frac{1}{2}\, E_{mat} A \left(\frac{\partial u}{\partial s}\right)^2 \tag{S11}$$

$$w_{bend}(s) = \frac{1}{2}\, E_{mat} I_{flex}\, \kappa^2(s) \tag{S12}$$

$$\kappa(s) = \frac{\partial \theta_{flex}}{\partial s} \tag{S13}$$

$$w_{reorient}(s) = \frac{1}{2}\, G_{mat} J\, \theta^2_{reorient}(s) \tag{S14}$$

Integrating these densities along the chain yields the three modal energies

$$\Psi_{debond} = \int_0^L w_{debond}(s)\, ds \tag{S15}$$

$$\Psi_{bend} = \int_0^L w_{bend}(s)\, ds \tag{S16}$$

$$\Psi_{reorient} = \int_0^L w_{reorient}(s)\, ds \tag{S17}$$

and the total mechanical energy

$$\Psi_{tot} = \Psi_{debond} + \Psi_{bend} + \Psi_{reorient} \tag{S18}$$

The debonding energy Ψ_{debond} measures how much work is stored in stretching and separating the material along the polymer-inspired chain axis in a backbone-controlled manner, and is therefore directly responsible for stiffness and strength. The bending energy Ψ_{bend} quantifies bending of the chain's path and describes how much strain can be accommodated by non-affine changes in shape before the chain enters a debonding-dominated regime. The segmental reorientation energy $\Psi_{reorient}$ measures local twist and reorientation and is the energetic counterpart of dihedral rotations and entanglement slip in real polymers. To compare different architectures and loading states, it is convenient to work with the dimensionless fractions

$$\Phi_{debond} = \frac{\Psi_{debond}}{\Psi_{tot}} \tag{S19}$$

$$\Phi_{bend} = \frac{\Psi_{bend}}{\Psi_{tot}} \tag{S20}$$

$$\Phi_{reorient} = \frac{\Psi_{reorient}}{\Psi_{tot}} \quad \text{(S21)}$$

which satisfy

$$\Phi_{debond} + \Phi_{bend} + \Phi_{reorient} = 1 \quad \text{(S22)}$$

These three fractions are computed from the numerical simulations by integrating the corresponding mode energies for each chain and averaging over the ensemble at a given global strain. The four deformation stages introduced in the main text can be rephrased in terms of the evolution of $\Psi_{debond}, \Psi_{bend}, \Psi_{reorient}$ as the macroscopic strain increases. In the segmental reorientation stage, the energy is dominated by $\Psi_{reorient}$, with Ψ_{bend} and Ψ_{debond} much smaller, so $\Phi_{reorient} \approx 1$. In the mixed reorientation–bending stage, Ψ_{bend} becomes comparable to $\Psi_{reorient}$, while Ψ_{debond} remains small; the fractions $\Phi_{reorient}$ and Φ_{bend} are both significant. In the onset of debonding-dominated loading, Ψ_{debond} grows and surpasses Ψ_{bend}, while $\Psi_{reorient}$ tends to saturate. Finally, in the debonding-dominated regime, Ψ_{debond} becomes the largest term by far, Φ_{debond} approaches unity, and the chain is close to its local failure.

The architected design space is parameterized by the polymer-inspired crosslink count per chain (ν_{cl}), the polymer-inspired entanglement count per chain (ν_{ent}), the polymer-inspired proto-crystalline ordering (Ω) that measures the proximity to a crystalline-inspired configuration, and a phantom connectivity factor. Here, the proto-crystalline chain refers to PIM chains simulating the pre-alignment caused by the mechanical effect of crystallization process of polymer. The simulations (Supplementary Figs. 26-30) reveal systematic and monotonic dependencies of the mode fractions on these parameters. Increasing crosslink density ν_{cl} shortens the effective free segments between junctions, stiffens both bending and twist at the nodes and forces the chain to deform more affinely. This effect appears as an increase in Φ_{debond} and Φ_{bend} and a reduction in $\Phi_{reorient}$. The system is thus driven towards an earlier and stronger debonding-dominated regime. Increasing the entanglement count per chain ν_{ent} has the opposite effect on the non-affine modes. Each entanglement introduces a location where chain segments can rotate and flex relative to each other, so adding entanglements increases Φ_{bend} and $\Phi_{reorient}$ and reduces Φ_{debond} at a given global strain. This extends the non-affine deformation stages, delays the growth of debonding energy, while decreases the fracture strain due to stress concentration of bended entangled polymer-inspired chains. Increasing the proto-crystalline ordering parameter Ω aligns the polymer-inspired chains with the load axis. As a result, the architecture enters a debonding-dominated response much earlier; Φ_{debond} rises sharply, while Φ_{bend} and $\Phi_{reorient}$ are suppressed. This produces a pronounced increase in stiffness and strength but disables large fracture strains. The three energy partitions are obtained from the simulation-measured redistribution of debonding, chain bending, and segmental reorientation under programmed crosslinking, entanglement, and proto-crystalline ordering. To represent these partition trends in a compact form, a normalized soft-competition relation is adopted:

$$\Phi_{debond} = \frac{\exp(M_{debond})}{\exp(M_{debond})+\exp(M_{bend})+\exp(M_{reorient})} \quad \text{(S23)}$$

$$\Phi_{bend} = \frac{\exp(M_{bend})}{\exp(M_{debond})+\exp(M_{bend})+\exp(M_{reorient})} \quad \text{(S24)}$$

$$\Phi_{reorient} = \frac{\exp(M_{reorient})}{\exp(M_{debond})+\exp(M_{bend})+\exp(M_{reorient})} \quad \text{(S25)}$$

which guarantees $0 < \Phi_i < 1$ and $\Phi_{reorient} + \Phi_{bend} + \Phi_{debond} = 1$. M_{debond}, M_{bend}, and $M_{reorient}$ denote the relative mode preferences inferred from the simulation-measured partition trends measured with ν_{ent}, ν_{cl}, Ω, f_{cl} given by Fig. 2. M_{debond} describes the tendency of the programmed chain architecture to store mechanical work through affine backbone-dominated stretching and separation, M_{bend} describes the tendency to accommodate work through non-affine chain curvature change and flexural deformation, and $M_{reorient}$ describes the tendency to accommodate work through segmental rotation and reorientation before the onset of debonding-dominated loading. These partitions are then related to the macroscopic properties and compared with classical polymer strengthening effects. To separate the primary effect of debonding from the weaker contributions of chain bending and reorientation, the following effective fraction is introduced

$$\Phi_E = \Phi_{debond} + k_E^{bend}\,\Phi_{bend} + k_E^{reorient}\,\Phi_{reorient} \quad \text{(S26)}$$

where k_E^{bend} and $k_E^{reorient}$ quantifies the relative efficiency of chain bending and segmental reorientation, respectively, in contributing to macroscopic stiffness compared with debonding-dominated backbone load transfer. This partition-based form links the present stiffness description to the strengthening relations, with debonding representing the dominant load-bearing contribution and bending/reorientation representing weaker non-affine contributions. Similar to classical polymer theory that modulus is proportional to the effective network chain density, the energy partition simulation indicated a relation

$$E = E_1\,\Phi_E^{m_E} \quad \text{(S27)}$$

Where E_1 is the constant modulus scale, and m_E capture the potential nonlinearity induced by approaching proto-crystalline or highly crosslinked states. The ultimate stress σ_c is governed by the fraction of energy carried in load paths that can sustain stress up to failure. An effective fraction Φ_σ is defined by

$$\Phi_\sigma = \Phi_{debond} + k_\sigma^{bend}\,\Phi_{bend} + k_\sigma^{reorient}\,\Phi_{reorient} \quad \text{(S28)}$$

where the coefficients reflect the level of polymer-inspired chains that chain-bending and reorientation energy distributions tend to engage at the strength point. The strength is then expressed as

$$\sigma_c = \sigma_1\,\Phi_\sigma^{m_\sigma} \quad \text{(S29)}$$

with σ_1 as the constant strength scale and m_σ an exponent that characterizes the degree of cooperative load sharing. Fracture strain ε_f is governed by how long the network can remain in regimes where non-affine modes carry a significant fraction of the energy before debonding dominates. ε_f is scaled inversely with debonding and negatively affected by bending, and enhance with reorientation modes:

$$\varepsilon_f = \varepsilon_1\,\Phi_{debond}^{-p}(1 + \chi_{reorient}\,\Phi_{reorient} + \chi_{bend}\,\Phi_{bend}) \quad \text{(S30)}$$

where ε_1 is the constant fracture strain scale. Exponent $p > 0$ controls the decay rate of fracture strain when the material becomes more debonding-dominated. The coefficients $\chi_{reorient}$ and χ_{bend} quantify how efficient segmental reorientation and chain-bending are

at delaying or encouraging fracture when their energy fractions are increased. The fracture energy is defined as

$$W_f = \int_0^{\varepsilon_f} \sigma(\varepsilon)\, d\varepsilon \tag{S31}$$

For the present networks, the stress–strain curves are well described by a family of shapes that differ mainly in their amplitude and horizontal scale, the fracture energy is then modelled as

$$W_f = W_1\, \sigma_c^{m_1}\, \varepsilon_f^{m_2} \tag{S32}$$

where W_1 sets the energy scale and m_1, m_2 are exponents of order one. When these exponents are determined from the data together with the other parameters, they are found to be close to unity, consistent with the physical picture that both sustained stress and strain window contribute comparably to the fracture energy.

The model parameters are derived from experimental measurements obtained through controlled single-parameter sweeps, in which each strengthening mechanism including crosslink density per chain, entanglement per chain, and proto-crystalline order is varied independently while the other parameters are fixed. Specifically, the crosslink density per chain is varied from 0 to 34 while the entanglement per chain is fixed at 10 and the proto-crystalline order is fixed at 0.57. The proto-crystalline order is then varied from 0.57 to 0.8 while the crosslink density remains zero and the entanglement per chain is fixed at 10. Similarly, the entanglement per chain is varied from 3.5 to 16.5 while the crosslink density is zero and the proto-crystalline order is fixed at 0.57. These controlled variations establish the baseline relationships between individual strengthening parameters and the corresponding energy partition responses. To validate the effectiveness of the proposed energy partition framework, held-out experiments are conducted on new samples that are not involved in model derivation and incorporate combined variations of crosslink density per chain, entanglement per chain, and proto-crystalline order. In these designs, the target energy partitions are prescribed through a design algorithm driven by the energy partition model as shown in Supplementary Fig. 3. Supplementary Fig. 31 compares the experimentally measured modulus, strength, fracture energy, and fracture strain with the model predictions, showing good agreement with a mean absolute percentage error below 10% across all measured properties.

Supplementary Note 7. Polymer-inspired strengthening mechanisms.

To validate the effectiveness of polymer-inspired strengthening mechanisms in PIMs. Experiments are conducted to compare the classical strengthening effects to PIMs designed with programmed toughening parameters including molecular density (MD) strengthening, Pre-stretched (PS) strengthening, and crosslink (CL) strengthening. According to the classical theory of polymer elasticity, the Young's modulus E of a polymer network is determined by the density of elastically effective network chains ν_e [2, 3]:

$$E = 3RT\nu_e \quad \text{(S33)}$$

where R is the gas constant and T is the absolute temperature. For networks containing both free molecular chains and chemical crosslinks, the effective network chain density is given by the superposition of contributions from these two components [4, 5]:

$$\nu_e = \rho + \beta n_c \quad \text{(S34)}$$

Here, ρ is the molecular chain density (number of chains per unit volume), n_c is the crosslink density (number of crosslinks per unit volume), and β is a constant related to the effectiveness of crosslinks in contributing to the network. According to the Lake–Thomas theory of fracture, the tensile strength σ_c of a polymer network scales with the square root of the effective network chain density [6]:

$$\sigma_c = \sigma_0\sqrt{\nu_e} \quad \text{(S35)}$$

where σ_0 is the reference strength. Pre-stretching aligns molecular chains along the loading direction, thereby modifying the mechanical response of the network. In oriented polymers, it is well established that both modulus and strength increase with the draw ratio λ [7]. According to the theory of chain orientation, the projected density of chain segments along the stretching direction increases, leading to a higher density of effectively load-bearing chains per unit volume, which enhances both modulus and strength [7]. Experimental data show that orientation strengthening follows a power-law relationship [8, 9]:

$$E = E_0\lambda^{n_E} \quad \text{(S36)}$$

$$\sigma_c = \sigma_0\lambda^{n_\sigma} \quad \text{(S37)}$$

where E_0 is the reference modulus, n_E and n_σ are positive exponents that reflect the sensitivity of modulus and strength to chain orientation, respectively. The draw ratio λ can be derived from:

$$\lambda = p/p_0 \quad \text{(S38)}$$

Here, p_0 denotes the pre-stretch level of the reference PIM to be strengthened, and p is the pre-stretch level of the strengthened PIM. The parameters of the formulas were determined by fitting to single-parameter sweep experimental data presented in Fig. 2. To validate the model, held-out tests (Supplementary Fig. 2) were performed on newly designed samples with composite strengthening parameters that were out of the range and values used in the calibration. The mean absolute percentage errors (MAPE) for the modulus and strength of these held-out samples range from 1.72% to 3.54% and from 0.95% to 3.55%, respectively, and consistent with predictions of energy partition models, which demonstrates that the PIMs can enable the relations from polymer-inspired strengthening effects.

Supplementary Note 8. Crystallization, compound crosslink-entanglement effect, and crosslink distributions of toughened PIMs.

In this section, we also investigated how crystallization, the competition between crosslinks and entanglements, and the homogeneity of the crosslink network modulate the strengthening of PIMs (designs and samples in Supplementary Fig. 32, corresponding mechanical results in Supplementary Fig. 33).

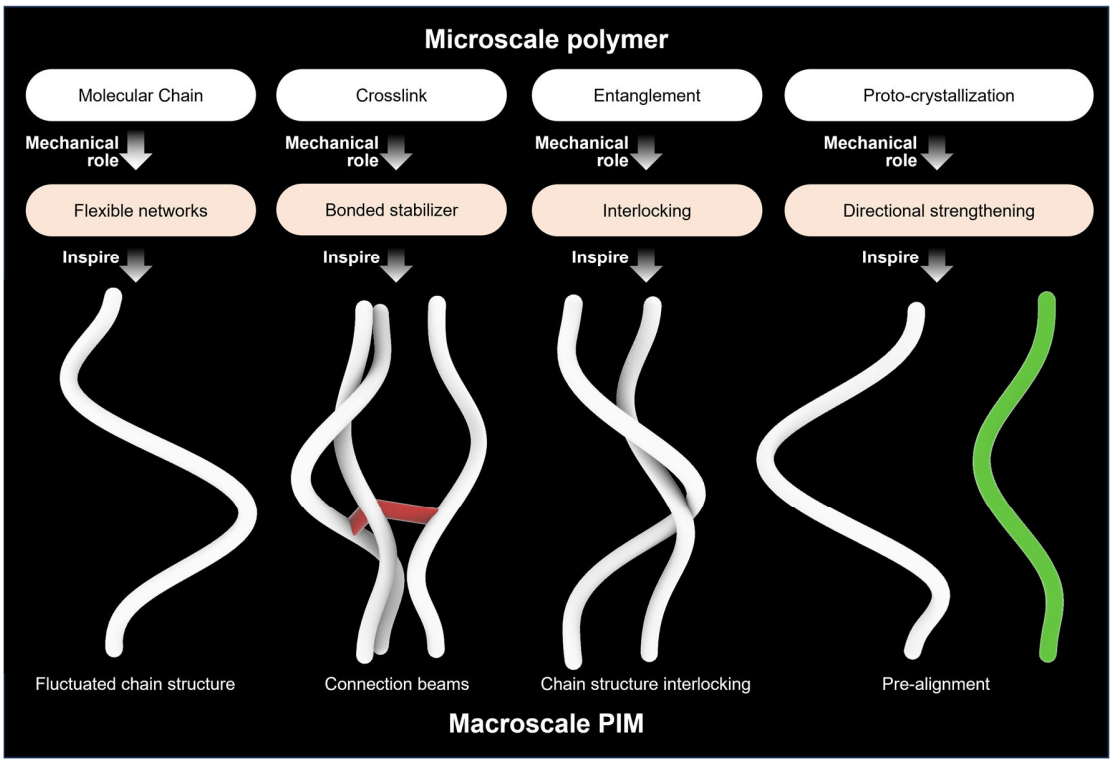
To probe the effect of crystallization beyond the proto-crystalline regime, we designed PIMs with different crystalline ratios χ_c from 0-100%. Each specimen contains a crystalline region flanked by maximally proto-crystalline regions on sides (Supplementary Fig. 32a). Here, the crystallization region refers to the space where PIM chains are designed to mimic the mechanics of closely aligned crystallized polymer chains. The experimental results in Supplementary Fig. 33a show that, as χ_c increases, the modulus exhibits a pronounced jump at the onset of crystallization between $\chi_c = 0\%$ and $\chi_c \approx 20\%$, where the PIM stiffens sharply, consistent with the debonding-energy partition predicted in Supplementary Fig. 30. While proto-crystalline ordering already increases the debonding contribution, true crystallization can drive the debonding partition Φ_{debond} effectively to unity, producing a large modulus increase. At full crystalline $\chi_c = 100\%$, the strength increases significantly, because the PIM no longer contains interfaces between crystalline and proto-crystalline regions. These observations mirror the classical behaviour of partially crystallized polymer networks, where the percolation of crystalline domains strongly constrains the amorphous matrix, and they provide a direct crystallization-based design guideline for PIMs.

In polymer mechanics, crosslinks are known to reduce extensibility unless entanglements dramatically outnumber crosslinks. To examine this interplay in PIMs, we designed highly entangled architectures in which each chain carries more than ten entanglements, and systematically programmed the ratio of crosslinks to entanglements ν_{cl}/ν_{ent} between 0 and 4% (Supplementary Fig. 32b). To quantify the combined influence to modulus (E) and fracture strain (ε_f), we defined an elastic–extensibility capacity $E\ \varepsilon_f$. As shown in Supplementary Fig. 33b, this capacity peaks at $\nu_{cl}/\nu_{ent} \approx 0.4\%$. At this point, chain-bending and debonding energies are both enhanced while the reorientation still shares a notable portion of partition. Beyond this optimum, increasing ν_{cl}/ν_{ent} makes the material progressively more brittle, the partition becomes debonding-dominated, and the elastic–extensibility capacity drops sharply. This identifies an optimal crosslink-to-entanglement ratio for highly entangled PIMs and is consistent with polymer-network concepts in which excessive crosslinking suppresses the beneficial non-affine modes provided by entanglements.

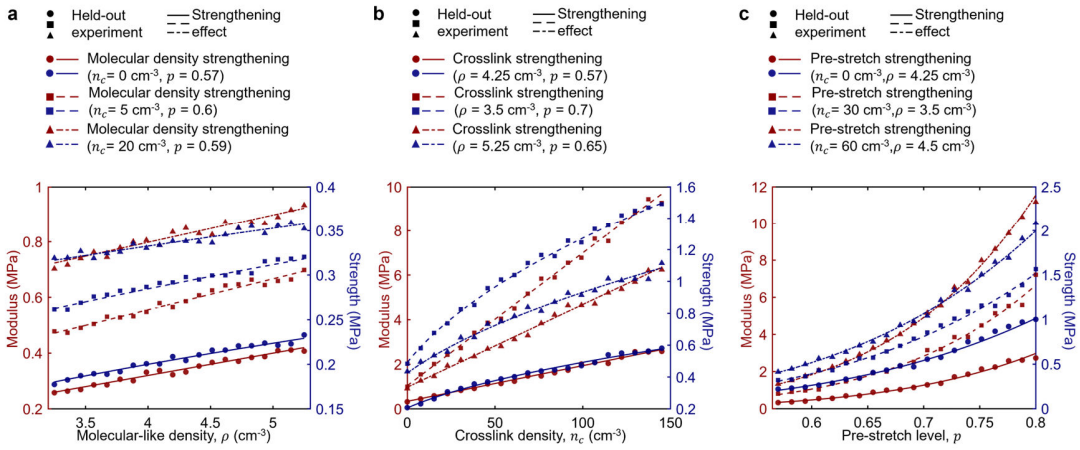
Since stress distribution in polymer networks is highly sensitive to crosslink homogeneity, we examined the effect of crosslink network uniformity in PIMs. We compared architectures with randomly distributed crosslinks and uniformly distributed crosslinks at nominally identical crosslink densities ranging from 25 to 100 cm^{-3} (Supplementary Fig. 32c). The designs reveal obvious clustering of crosslinks in the random case. The mechanical results in Supplementary Fig. 33c show that, at low crosslink

density, random crosslinking reduces the modulus by 13% compared with the uniform network. As the crosslink density increases to 100 cm^{-3}, this difference shrinks to 0.8%. This convergence arises because the random crosslinks become effectively more homogeneous at higher density, approaching the uniform distribution. These experiments therefore clarify how both the density and the spatial distribution of crosslinks control PIM performance, providing practical guidance for designing crosslink-based strengthening strategies in PIMs.

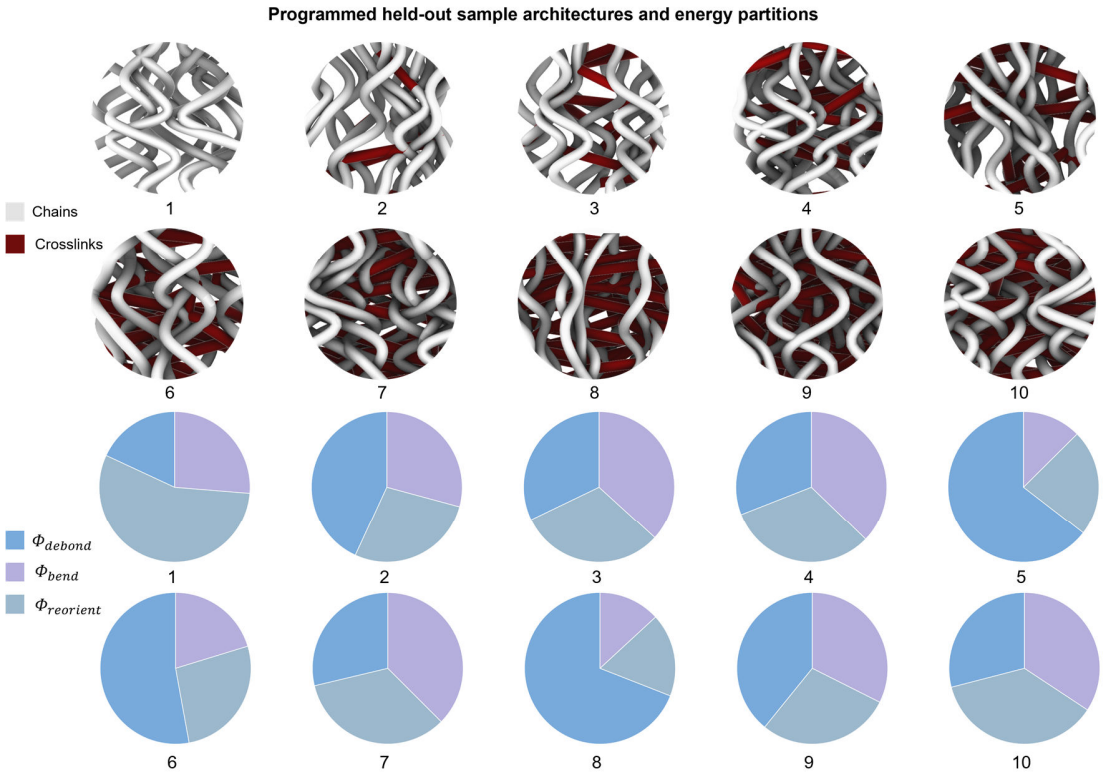
Supplementary figures



Supplementary Fig. 1. Macroscale PIMs generated by PSG algorithm inspired from the mechanical roles of microscale polymer.



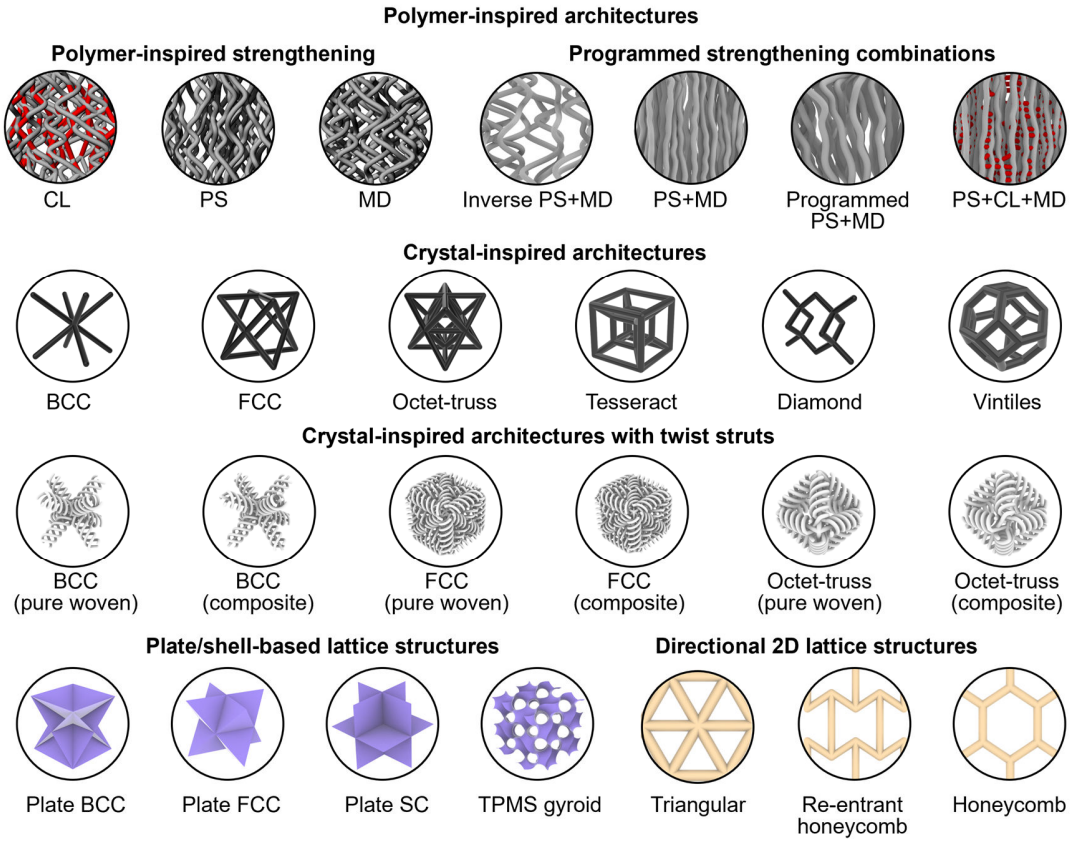
Supplementary Fig. 2. Experimental measurements of PIM strengthening compared with predictions derived from polymer strengthening mechanisms. This includes (a) molecular-density strengthening, (b) CL strengthening, and (c) PS strengthening effects. The strengthening relationships are established by equations S33-38 using parameter studies in Fig. 2 (ρ = 3.25-5.25 cm^{-3} at fixed n_c and p; n_c = 0-145 cm-3 at fixed ρ and p; p = 0.57-0.8 at fixed ρ and n_c). Independent held-out experiment validation is conducted on combined strengthening parameters that were not used in constructing these relationships. Source data are provided as a Source Data file.



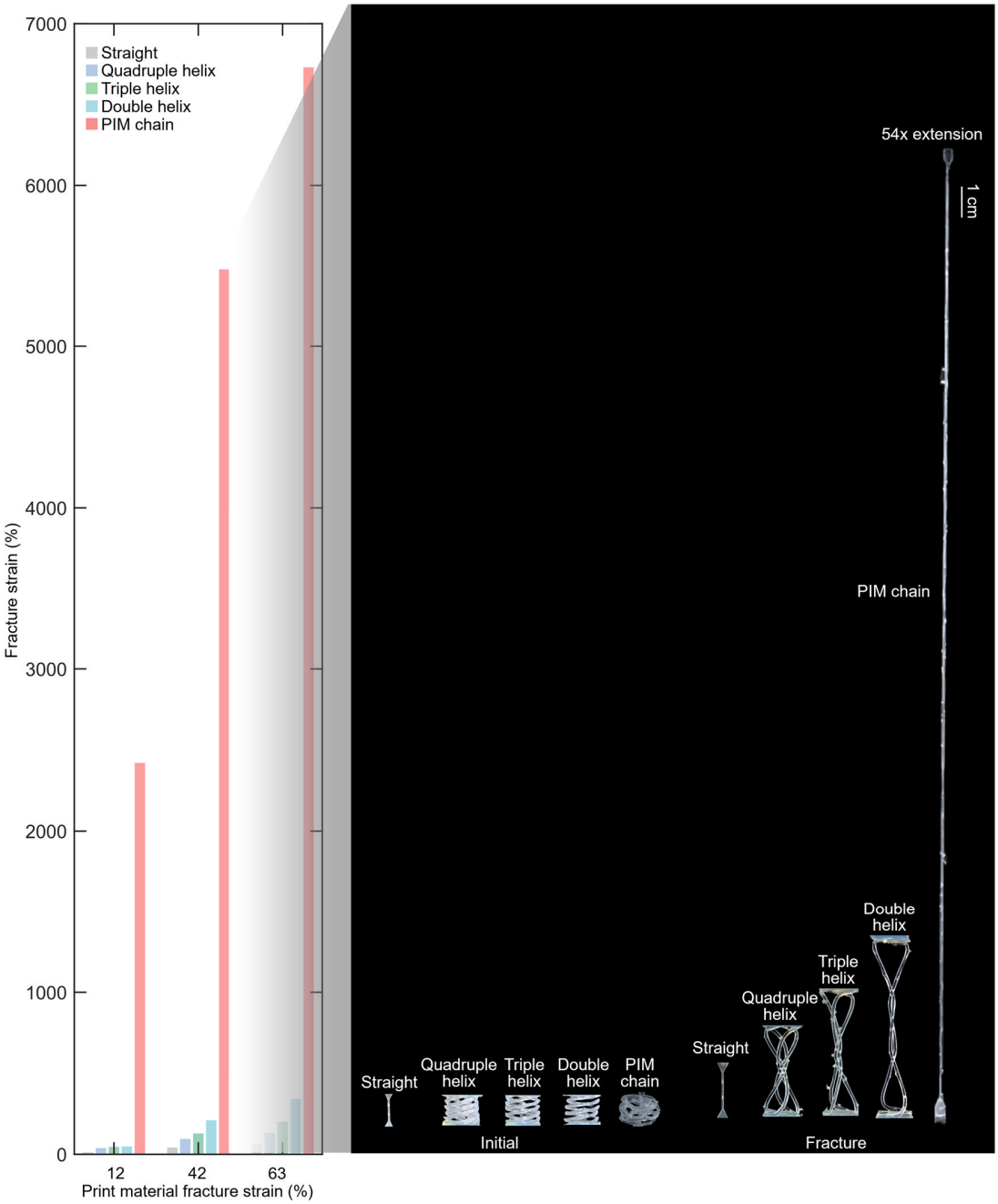
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529 Supplementary Fig. 3. Architected held-out PIM samples with diverse structural
530 configurations. These samples are programmed with different debonding (Φ_{debond}),
531 bending (Φ_{bend}), and reorientation ($\Phi_{reorient}$) energy partitions. Note that the PIM designs
532 presented here involve composite strengthening parameters that fall outside the parameter
533 values and single ranges used in Fig. 2. Source data are provided as a Source Data file.

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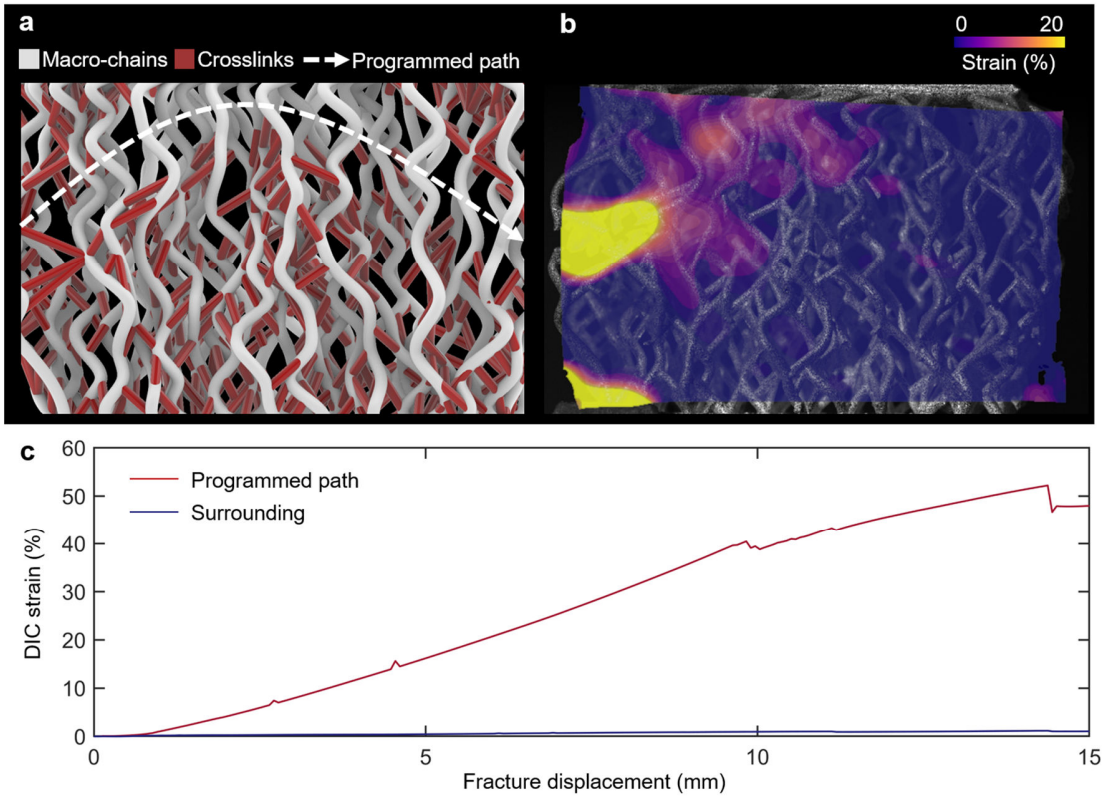


Supplementary Fig. 4. Structural designs. The topological designs of CIMs, CIMs with twist structures, plate/shell-based lattice structures, directional lattice structures, and PIMs with different strengthening mechanisms, as shown in Figure 3. Here, MD, PS, and CL denote MD strengthening, PS strengthening, and CL strengthening, respectively. The programmed strengthening combinations include inverse PS+MD, PS+MD, programmed PS+MD, and PS+CL+MD. BCC, FCC, and SC represent body-centered cubic, face-centered cubic, and simple cubic architectures, respectively. The twist-structured CIMs include BCC, FCC, and octet-truss configurations in both pure woven and composite forms. Plate/shell-based lattice structures include plate BCC, plate FCC, and plate SC, as well as TPMS gyroid structures. Directional lattice structures include triangular, re-entrant honeycomb, and honeycomb architectures.

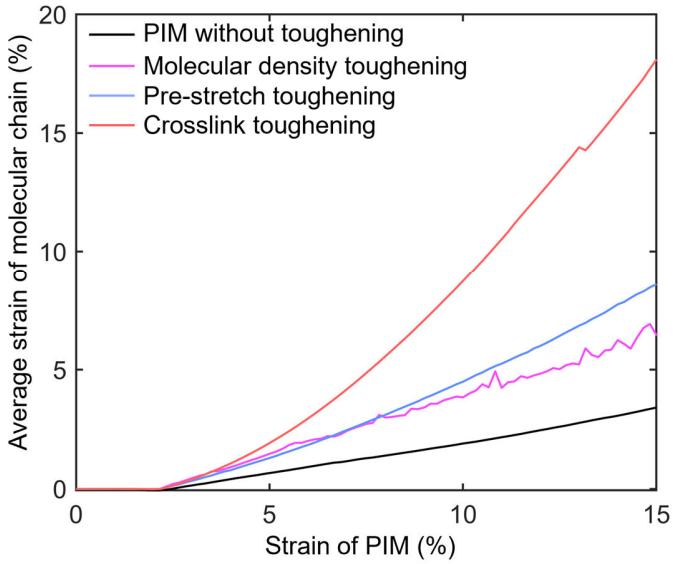


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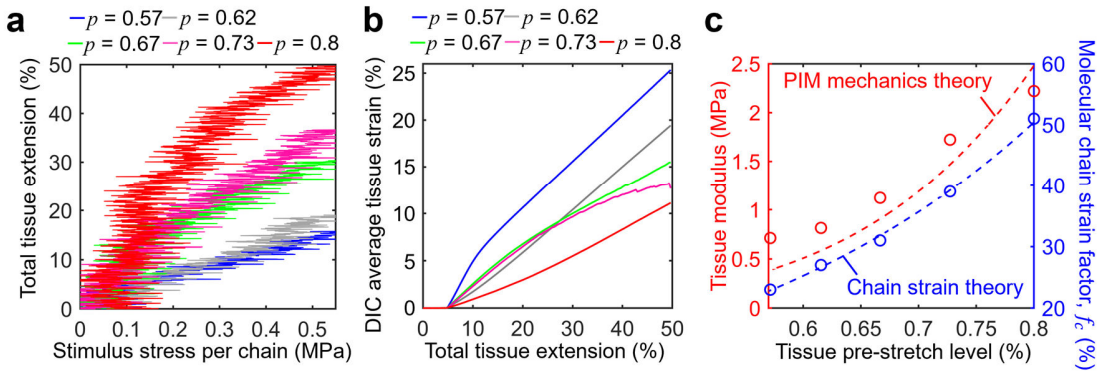
549 Supplementary Fig. 5. The fracture strains comparison at strut scale. The comparisons
550 include conventional straight struts, quadruple helix, triple helix, double helix, and PIM
551 chain at identical strut diameter and design space with fracture strains at ε_f = 12%, ε_f =
552 42%, and ε_f = 63%, respectively. Source data are provided as a Source Data file.
553



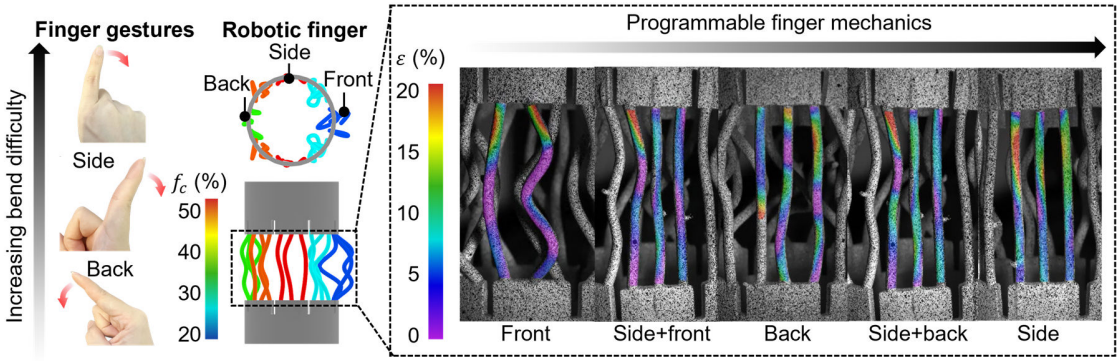
Supplementary Fig. 6. Crack path programming via artificial strengthening engineering of PIM. (a) A PIM with programmed crack path controlled by crosslink distributions, where the $n_c = 0$ and $n_c = 145$ cm^{-3} are applied at desired path and surroundings, respectively. (b) The DIC strain distribution at 3.5 mm fracture displacement. (c) DIC strains calculated from 20 random points located at the programmed path and surrounding. Source data are provided as a Source Data file.



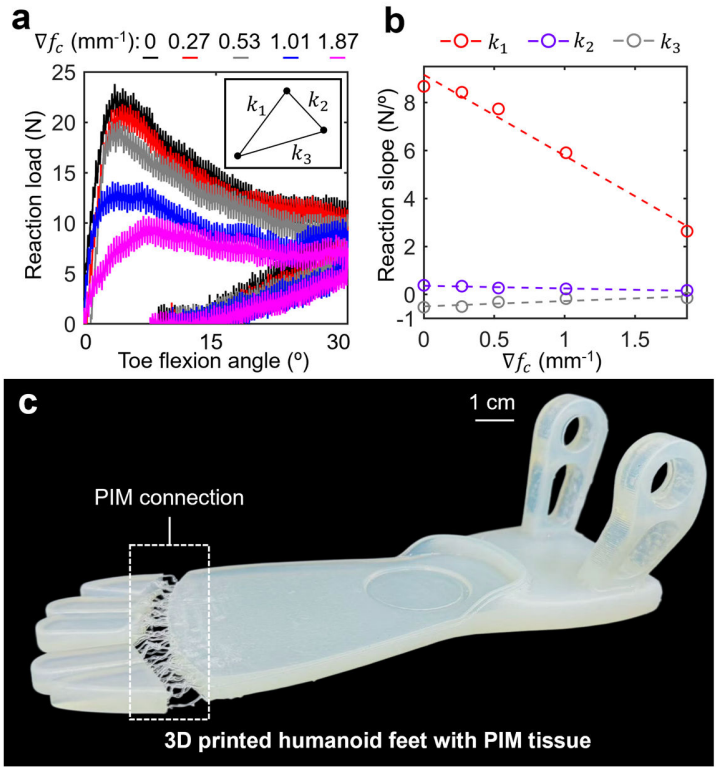
Supplementary Fig. 7. The chain strain averaged on the three molecular chains during the deformation processes of different types of PIMs. Source data are provided as a Source Data file.



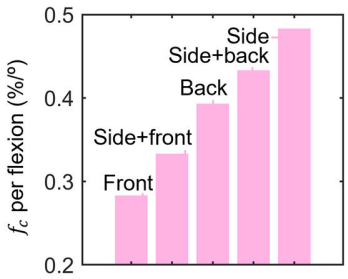
Supplementary Fig. 8. PIM mechanics and chain strain theories based on experimental DIC tissue stimulation measurement. (a) Programmable tissue extensions per stress stimulus for the PIM humanoid tissues. (b) The average strains on the polymer chains of the PIM humanoid tissues per tissue extension. (c) The tissue modulus and the molecular chain factor, f_c, of the PIM humanoid tissues programmed with different proto-crystalline levels. Source data are provided as a Source Data file.



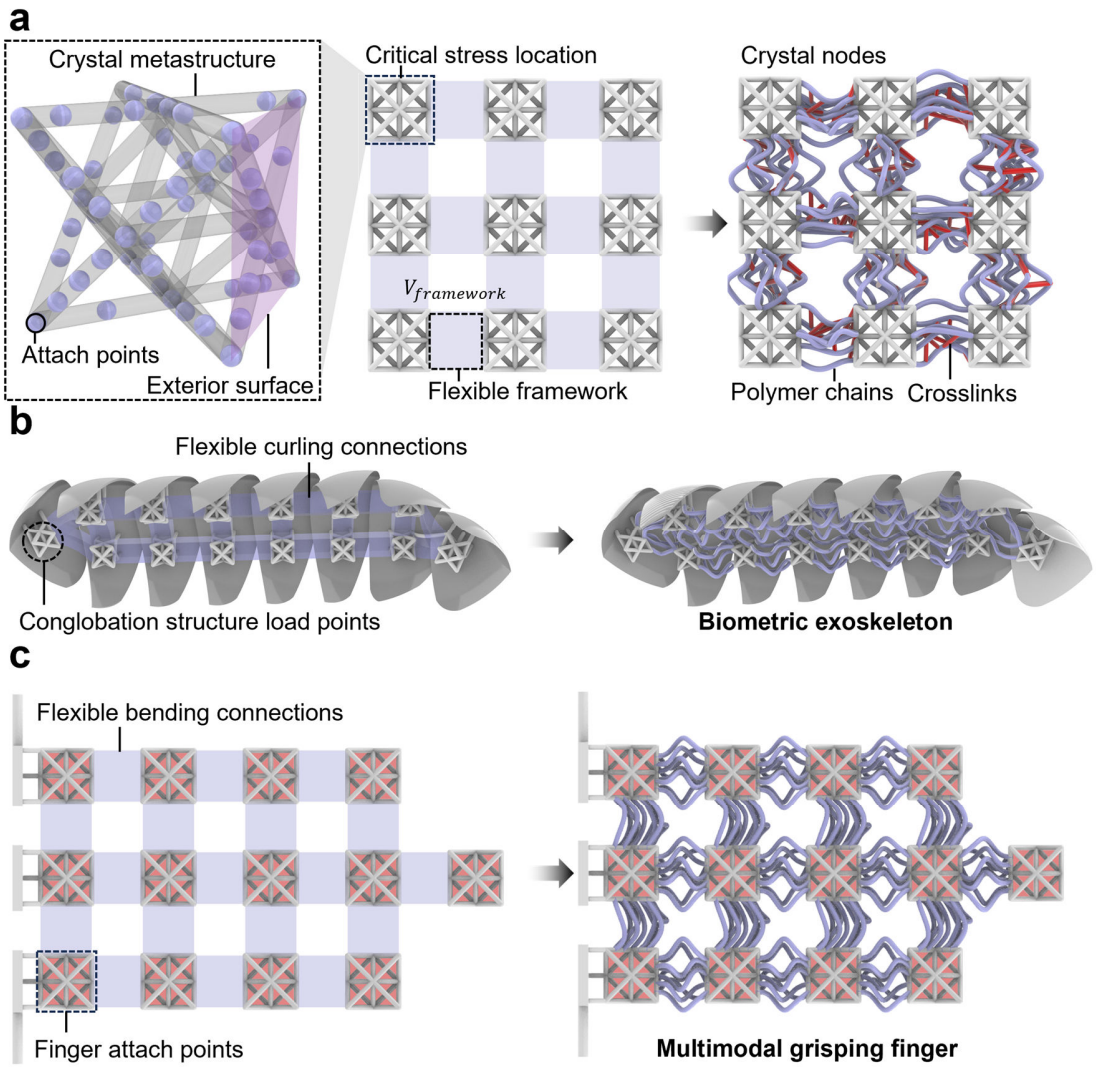
Supplementary Fig. 9. Experimental data of robotic finger. The robotic finger simulating the bending mechanics of human finger and the DIC analysis of the strain fields for different gestures of the robotic finger. The hand photos were photographed by author Zhenyang Gao.



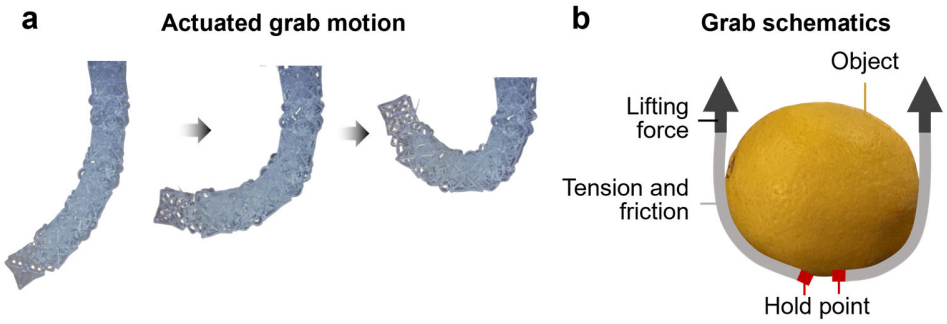
Supplementary Fig. 10. Experimental data and fabricated robotic foot. The (a) customized reaction load of the robotic foot with identical walking mechanics of human foot, (b) the programmable reaction slope data of the robotic foot, (c) a 3D printed humanoid robotic feet with PIM tissue. Source data are provided as a Source Data file.



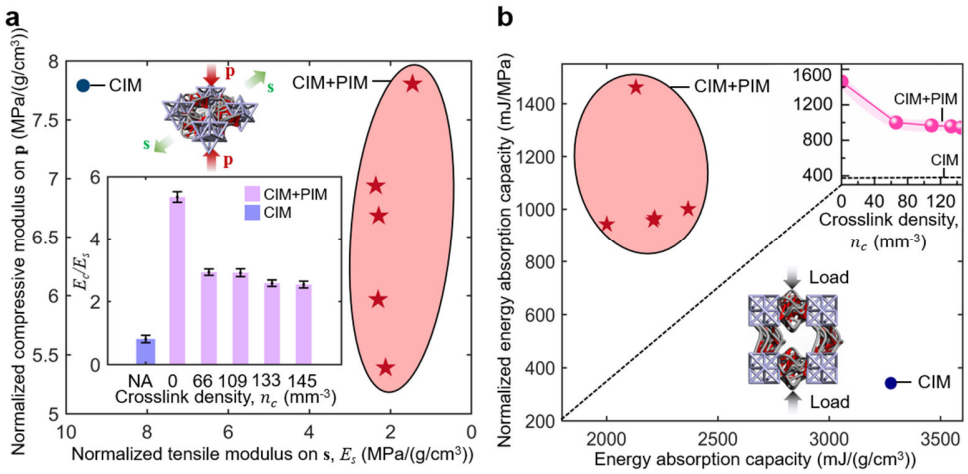
Supplementary Fig. 11. The programmable experimental molecular chain factor per flexion for the robotic finger. Source data are provided as a Source Data file.



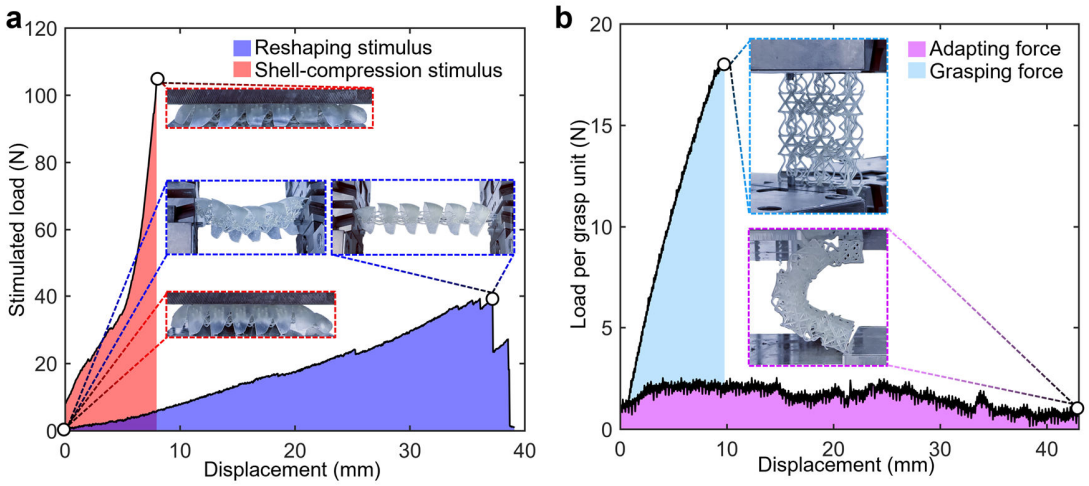
Supplementary Fig. 12. The CIM+PIM functional part designs. The (a) design schematics of CIM+PIM composites, (b) the design of a biometric exoskeleton enabled by CIM+PIM, and (c) the design of a multimodal robotic finger enabled by CIM+PIM.



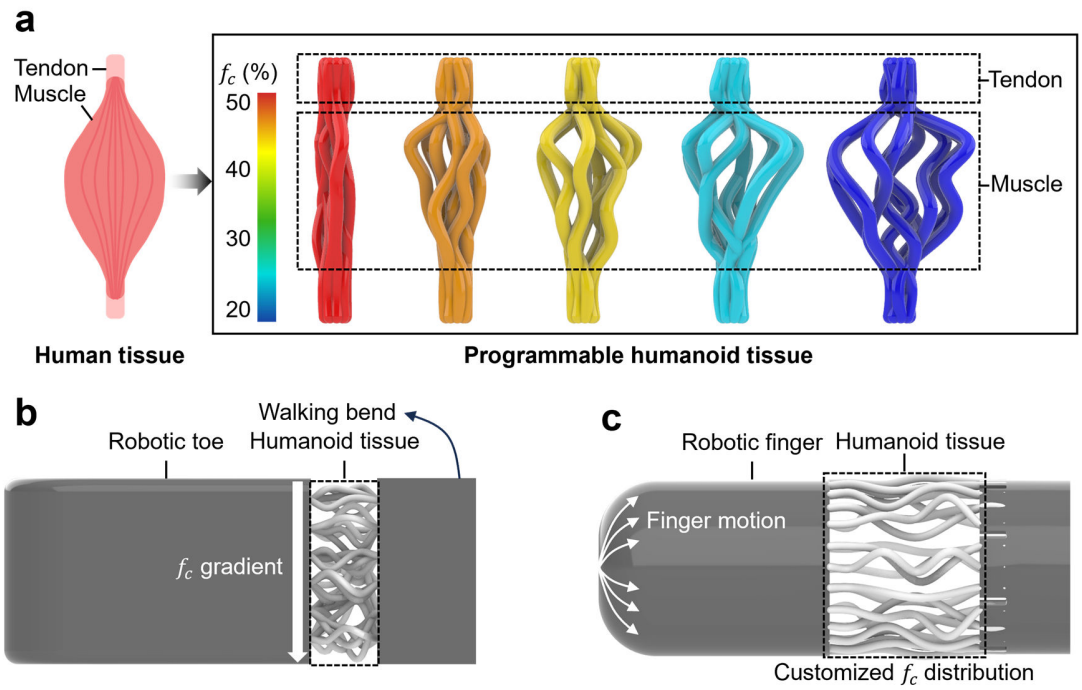
Supplementary Fig. 13. Schematics of actuated grab. The (a) cable-actuated grab motion and (b) schematics of grab mechanics.



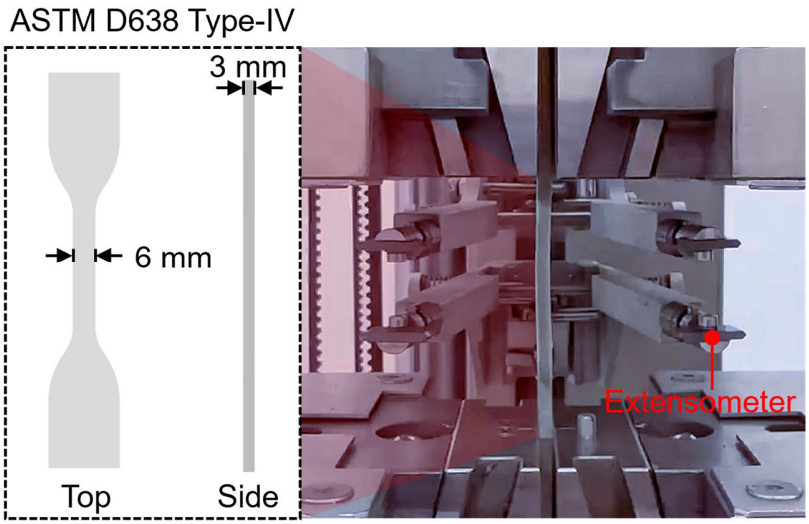
Supplementary Fig. 14. The anisotropic modulus and energy absorption capacities of CIM+PIM and CIM at different directions. (a) The anisotropic mechanical properties of CIM+PIM metamaterials compared to existing CIMs, where **p** and **s** are the directions of the crystal loading and polymer chain loadings specified in Methods, respectively. The error bars are calculated based on the standard deviations from the results of three repetitive experiments. (b) The energy absorption capacities normalized by the modulus values for CIM+PIM and CIM, there the modulus of CIM+PIM is along the direction of polymer chain loading. Source data are provided as a Source Data file.



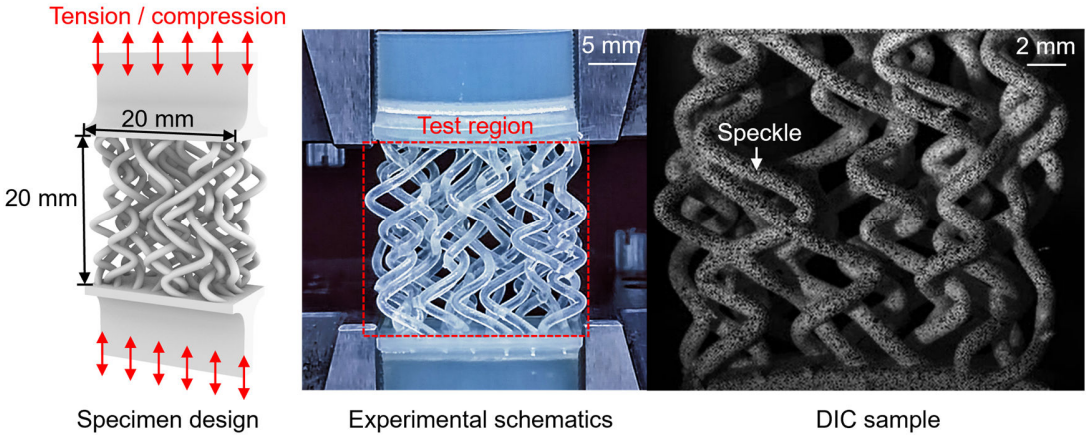
Supplementary Fig. 15. Mechanical evaluations of biometric exoskeleton and multimodal grab finger. (a) The stimulated shell-compression and reshaping loads of the biometric exoskeleton with CIM+PIM. (b) The adapting and grasping forces of the multimodal grab finger with CIM+PIM. Source data are provided as a Source Data file.



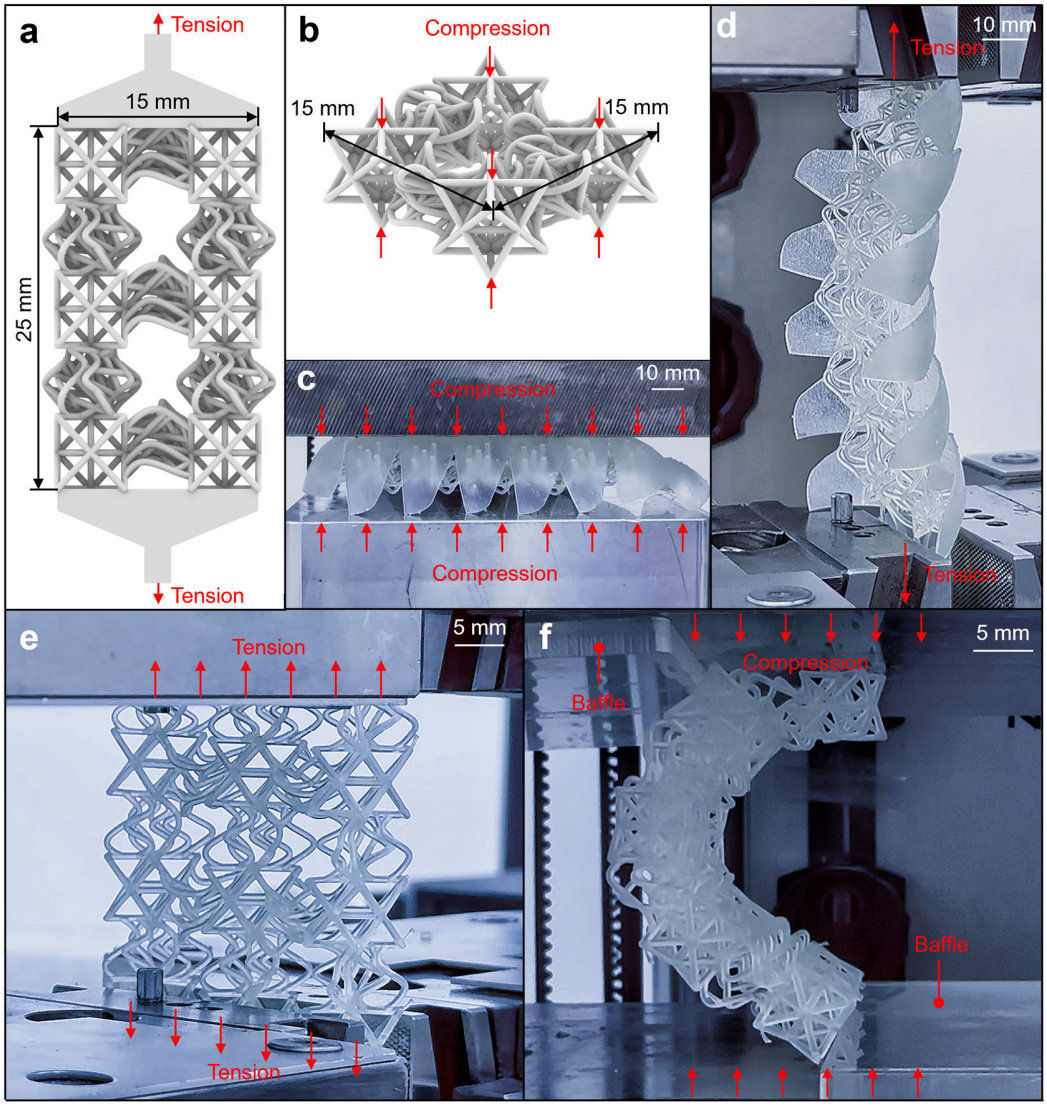
Supplementary Fig. 16. Programmable tissue and robotics designs. The (a) design of programmable humanoid tissues, and (b-c) the use of humanoid tissues in robotic toe and finger, respectively.



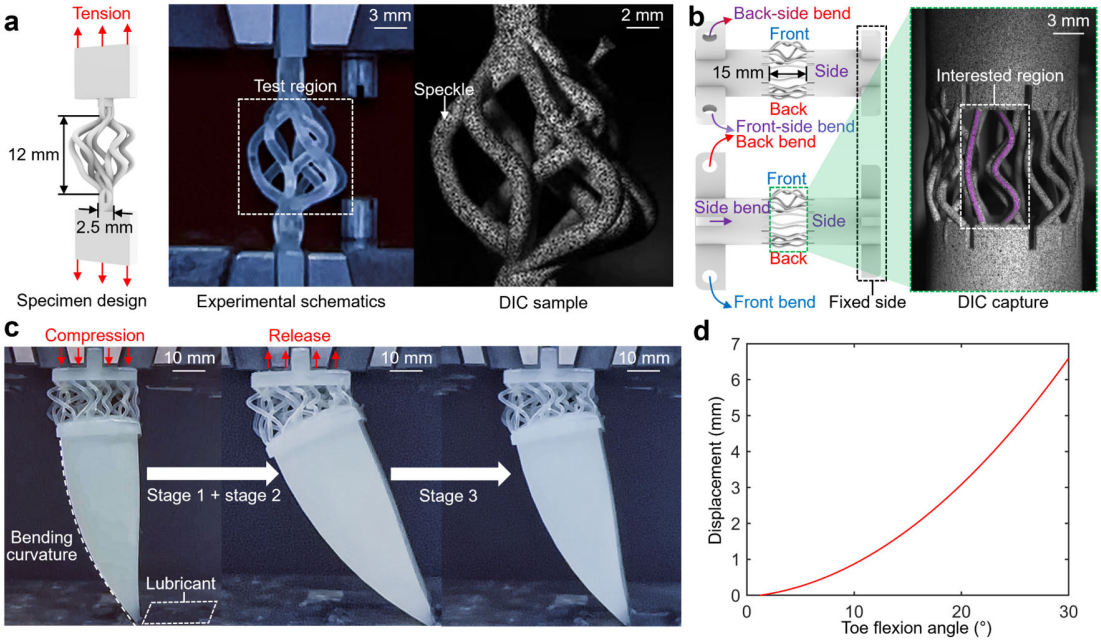
Supplementary Fig. 17. The mechanical test configurations for parent material evaluation.



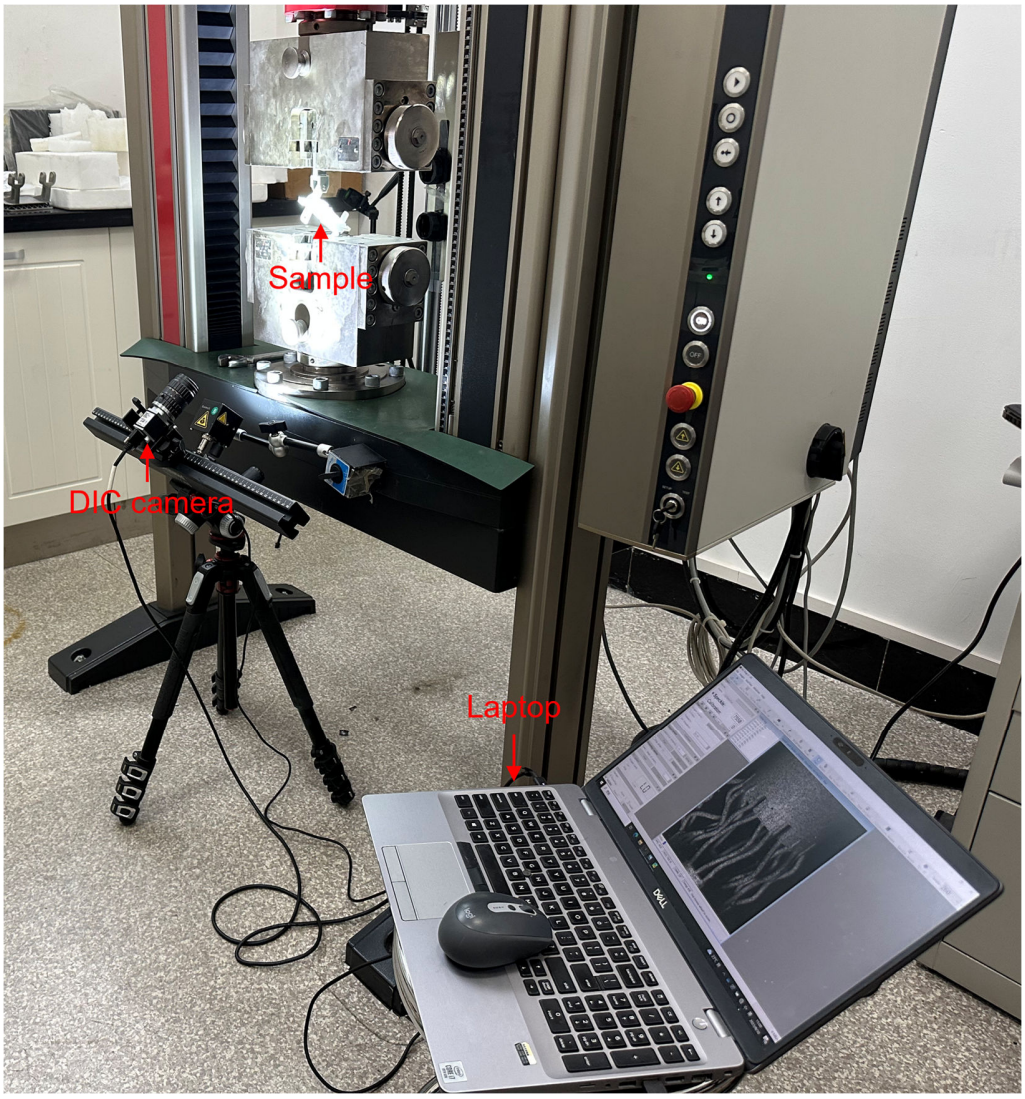
Supplementary Fig. 18. The mechanical test configurations for PIMs and CIMs with different strengthening mechanisms.



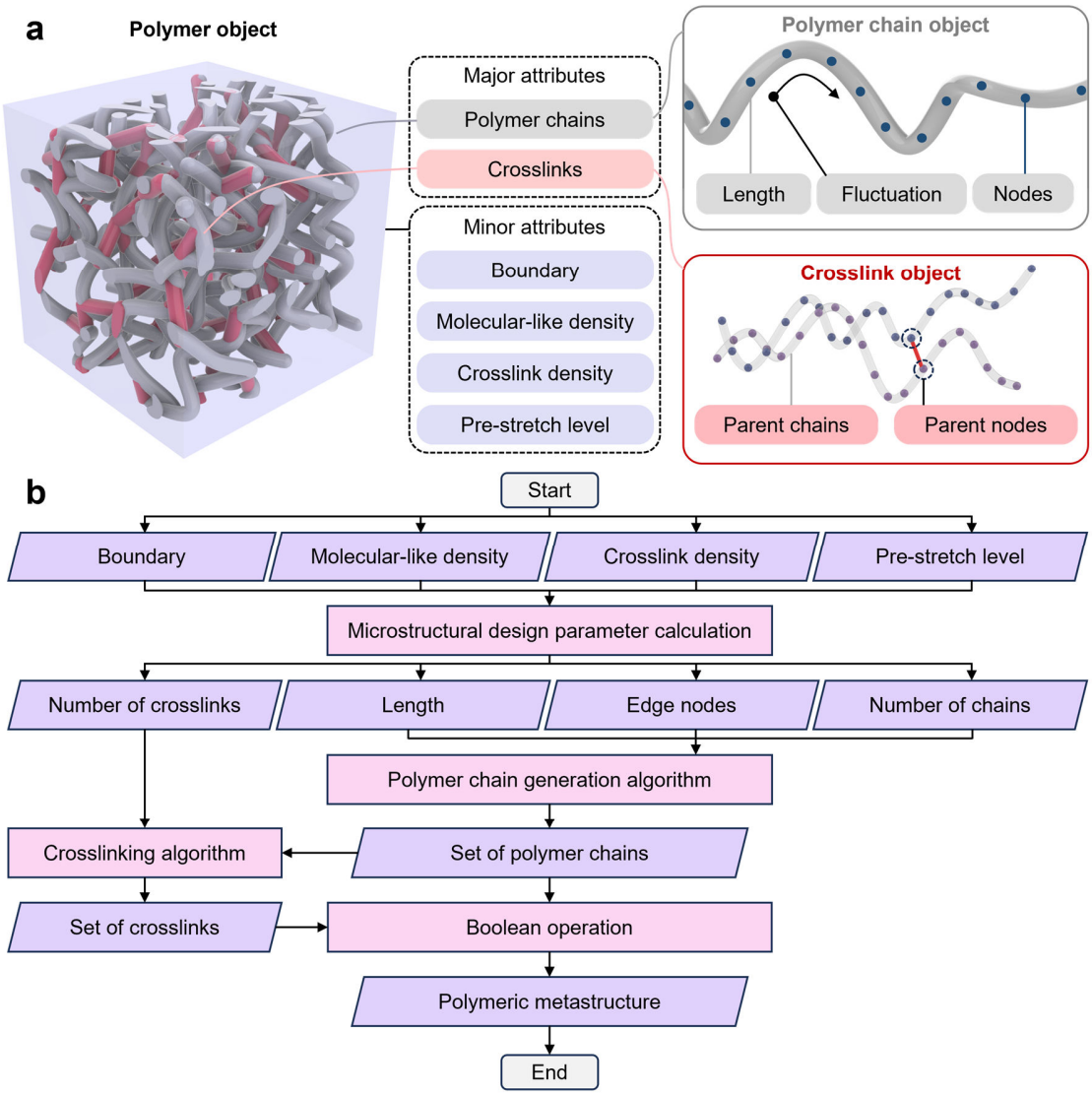
Supplementary Fig. 19. The mechanical test configurations for PIM+CIM composites. (a) Tensile experimental configuration measuring the mechanics along the polymer chain loading direction. (b) Compressive experimental configuration measuring the mechanics along the crystal loading direction. (c-d) Compressive and tensile loading of the biometric exoskeletons measuring its stimulated load on shell-compression and reshaping deformations, respectively. (e-f) Tensile and compressive-bending load scenarios for a single multimodal robotic finger, respectively.



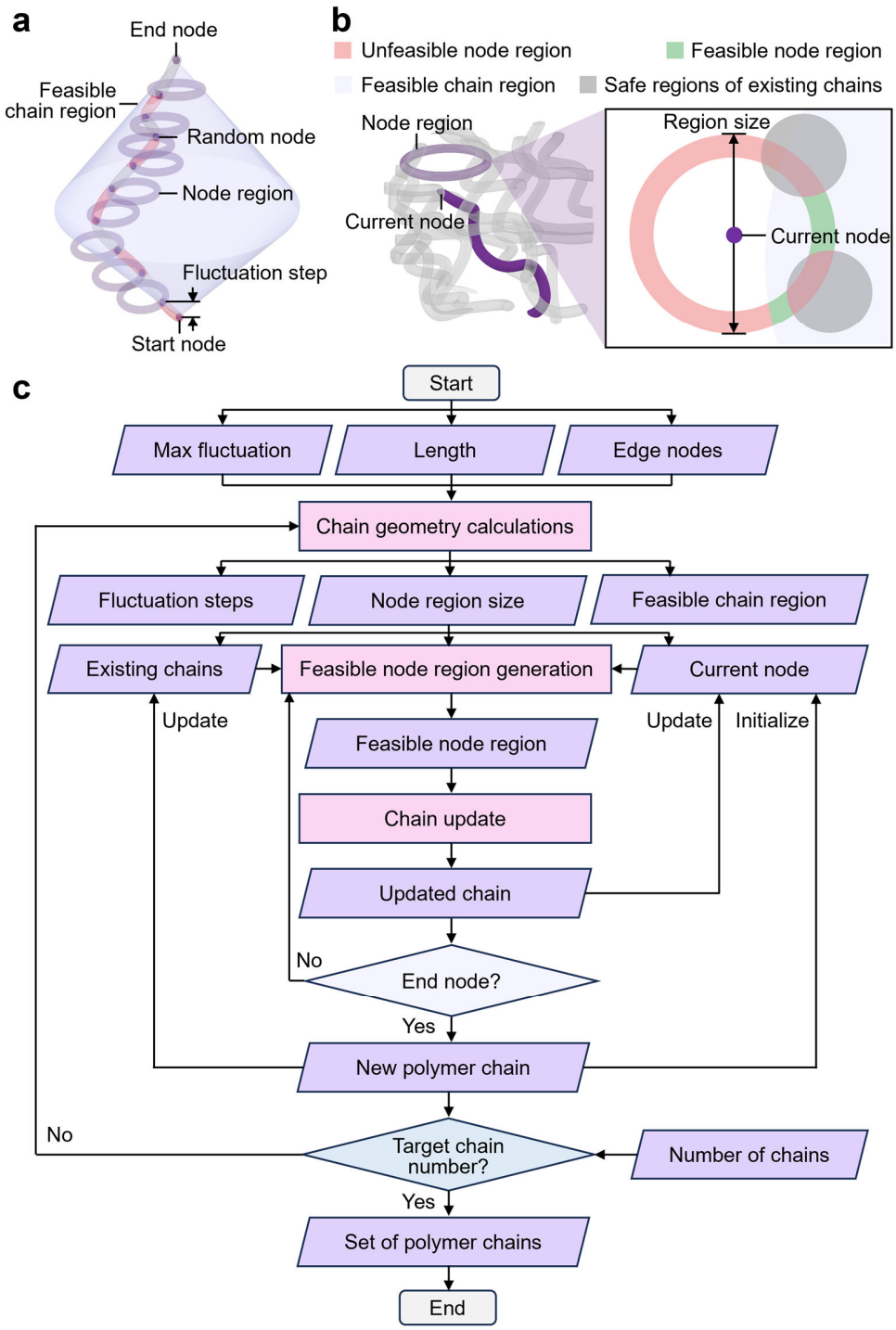
Supplementary Fig. 20. The mechanical test configurations for humanoid tissues. (a) Tensile test configuration for single humanoid tissue sample. (b) the bending experimental scenario for robotic finger tissues at different relative positions of the finger. (c) A compression and release cycle simulating a walking mechanics of human toe. (d) The relation between the displacement and toe flexion angle resulted from the bending curvature. Source data are provided as a Source Data file.



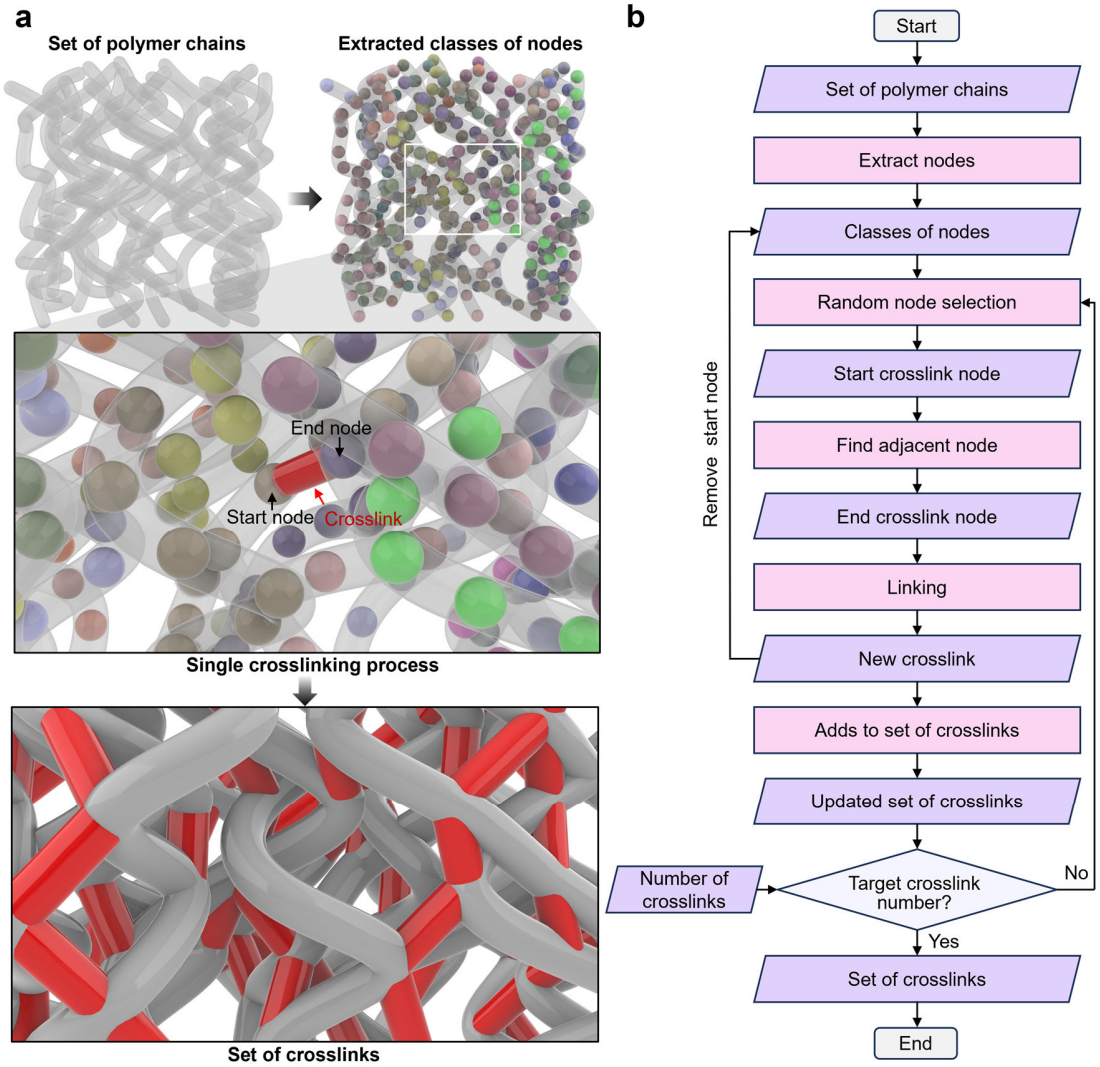
Supplementary Fig. 21. The experimental schematics and digital image correlation system.



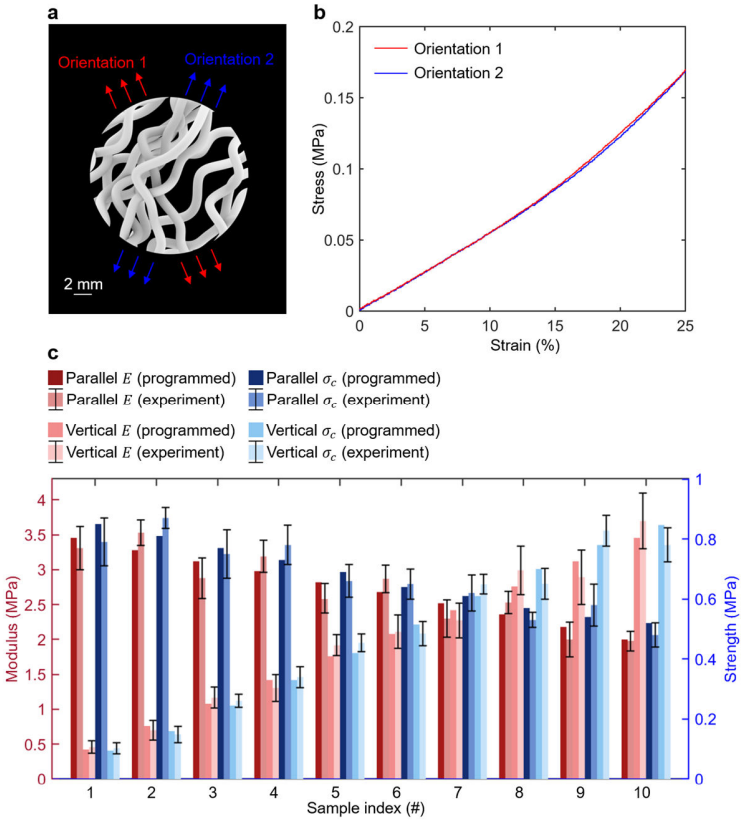
Supplementary Fig. 22. The polymeric meta-structure generation algorithm. (a) The object-oriented design schematics of a polymeric meta-structure. (b) The algorithmic design workflow of a polymeric meta-structure.



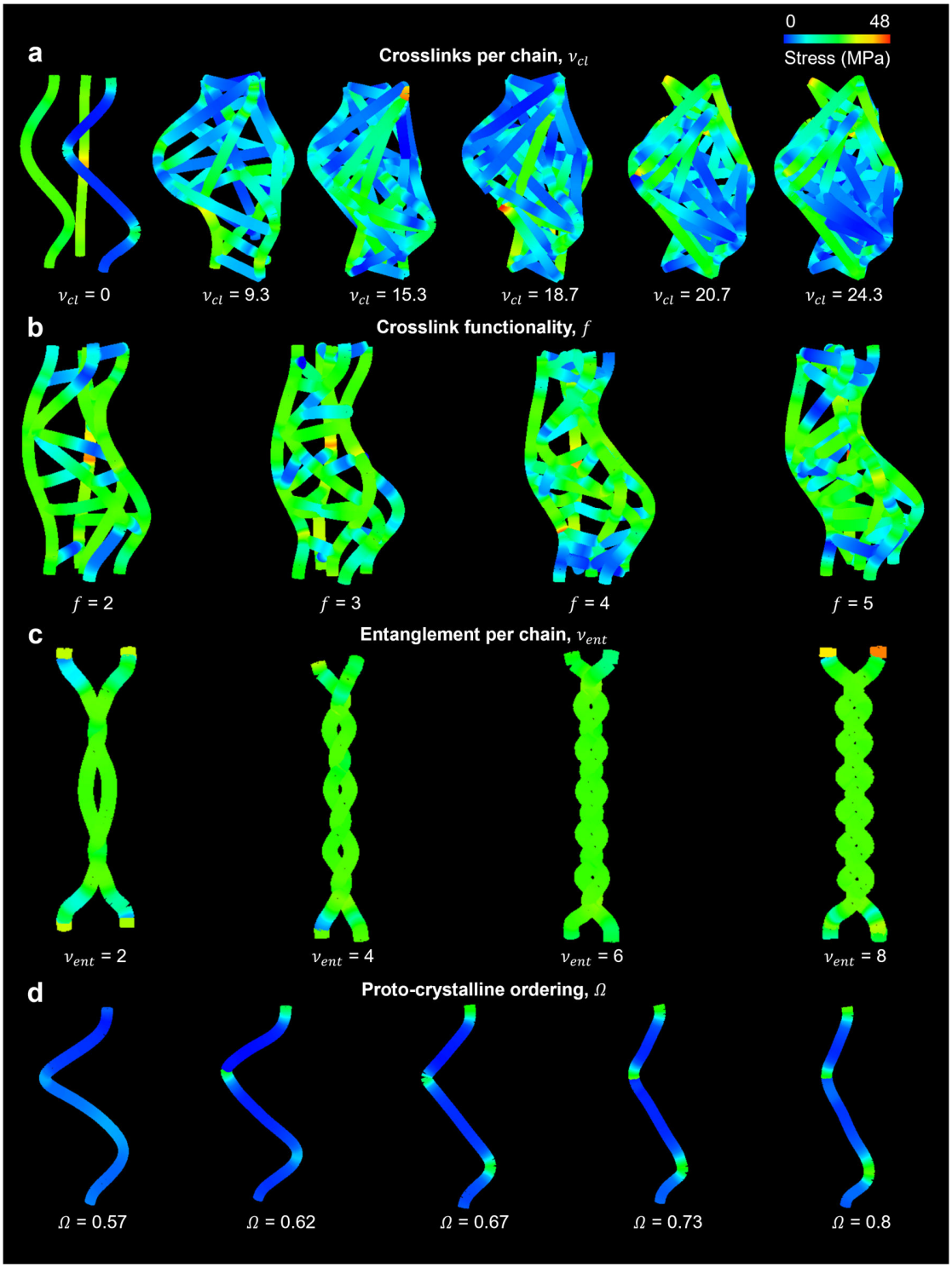
Supplementary Fig. 23. The polymer chain generation algorithm. (a-b) The design schematics of the chain geometry calculation and the feasible node region generation, respectively. (c) The algorithmic design workflow of a polymer chain.



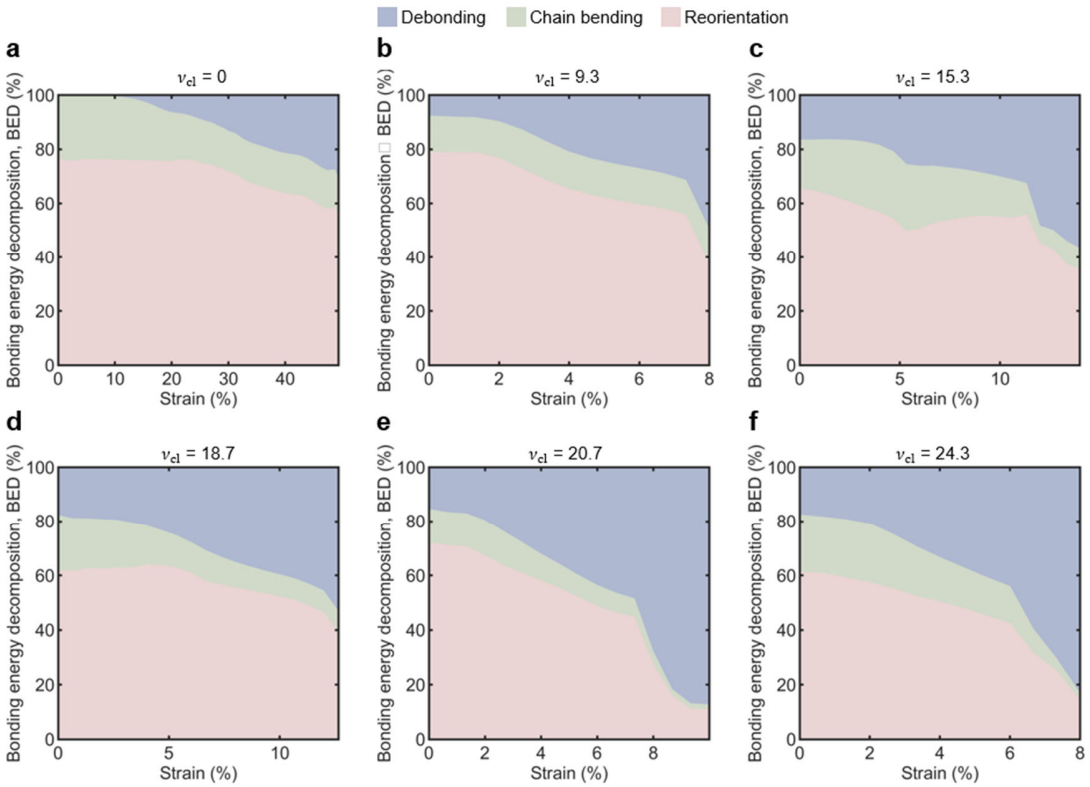
Supplementary Fig. 24. Crosslinking design. The (a) design schematics and (b) algorithmic design workflow of crosslinking algorithm.



Supplementary Fig. 25. Programmed bi-directionally isotropic PIM. (a) Microstructures of bi-directionally isotropic PIM, along with the two corresponding directions. Both directions are programmed with p = 57%. (b) Tensile stress–strain curves measured along the two programmed isotropic directions. (c) Programmed modulus and strength of 10 samples experimentally measured in the parallel and perpendicular load directions. The error bars are calculated based on the standard deviations from the results of three repetitive experiments. Source data are provided as a Source Data file.



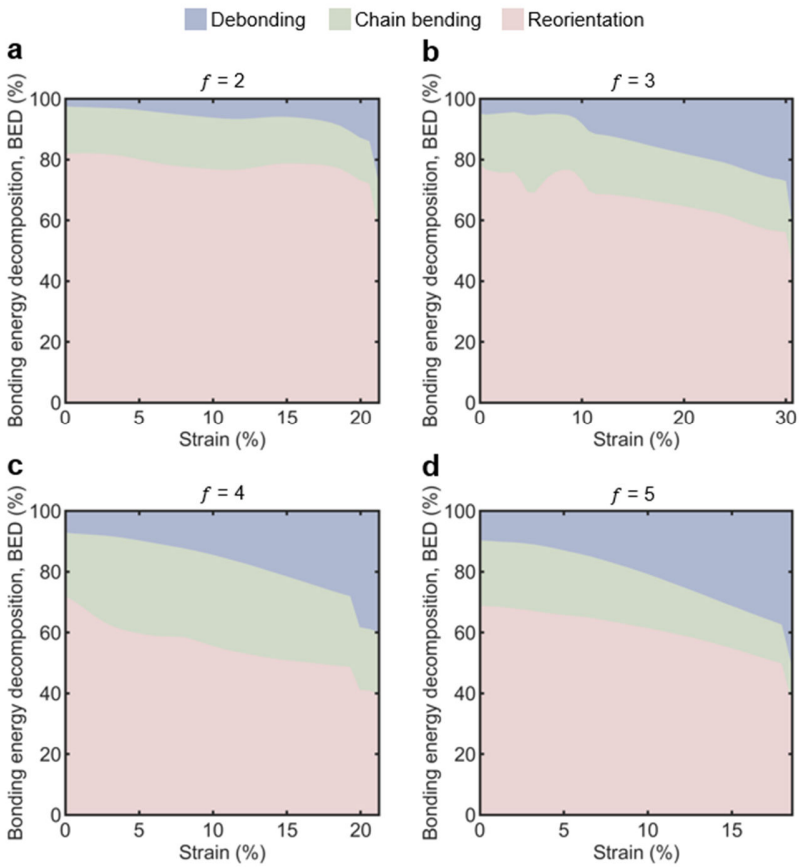
Supplementary Fig. 26. Finite element analysis of strengthening events. These events are triggered by (a) crosslinks per chain, (b) crosslink functionality, (c) entanglement per chain, and (d) proto-crystalline ordering.



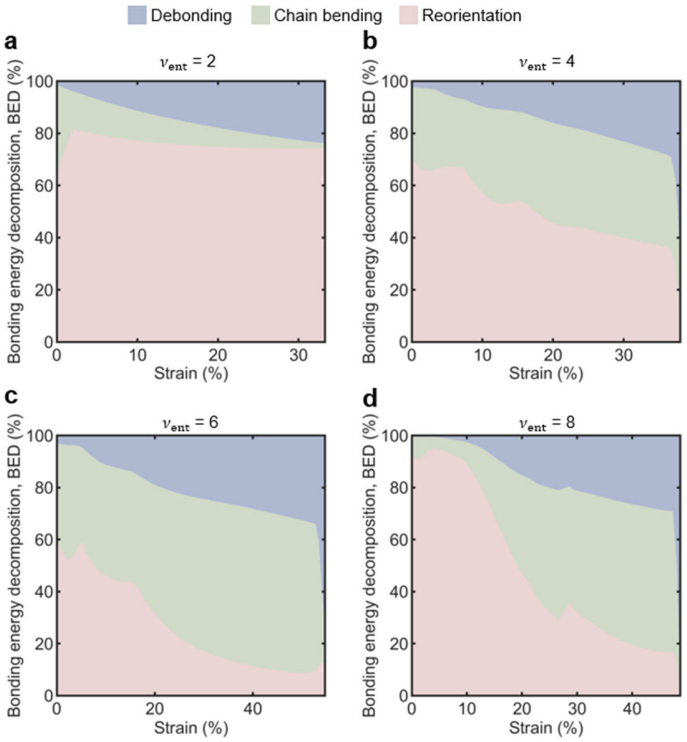
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674 Supplementary Fig. 27. Chain energy partition under different crosslinks per chain, ν_{cl}.
675 Source data are provided as a Source Data file.

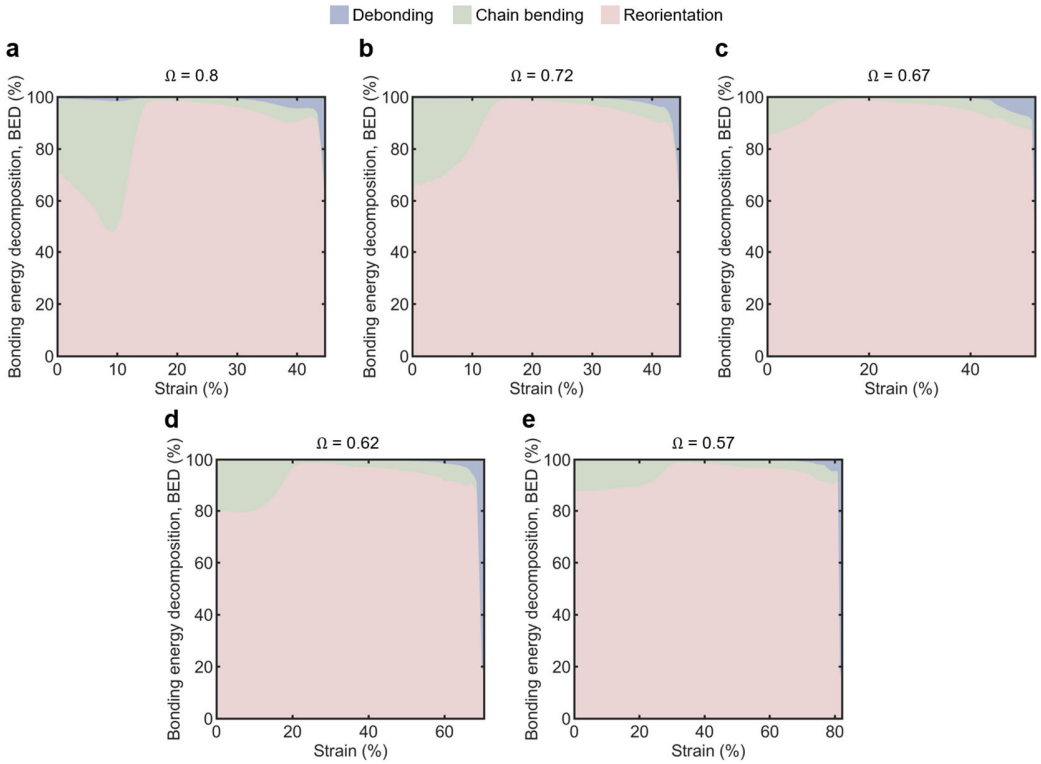
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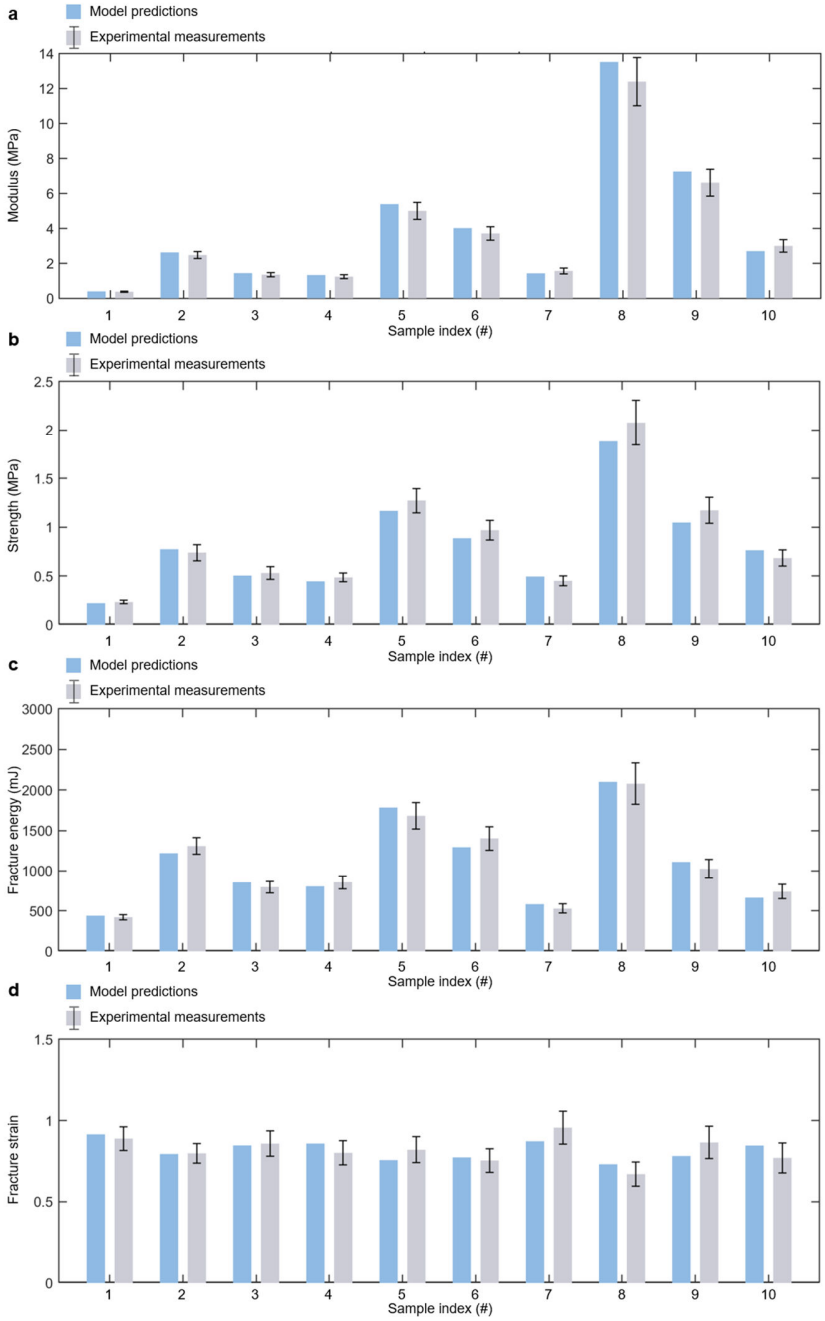
Supplementary Fig. 28. Chain energy partition under different crosslink functionality, f. Source data are provided as a Source Data file.



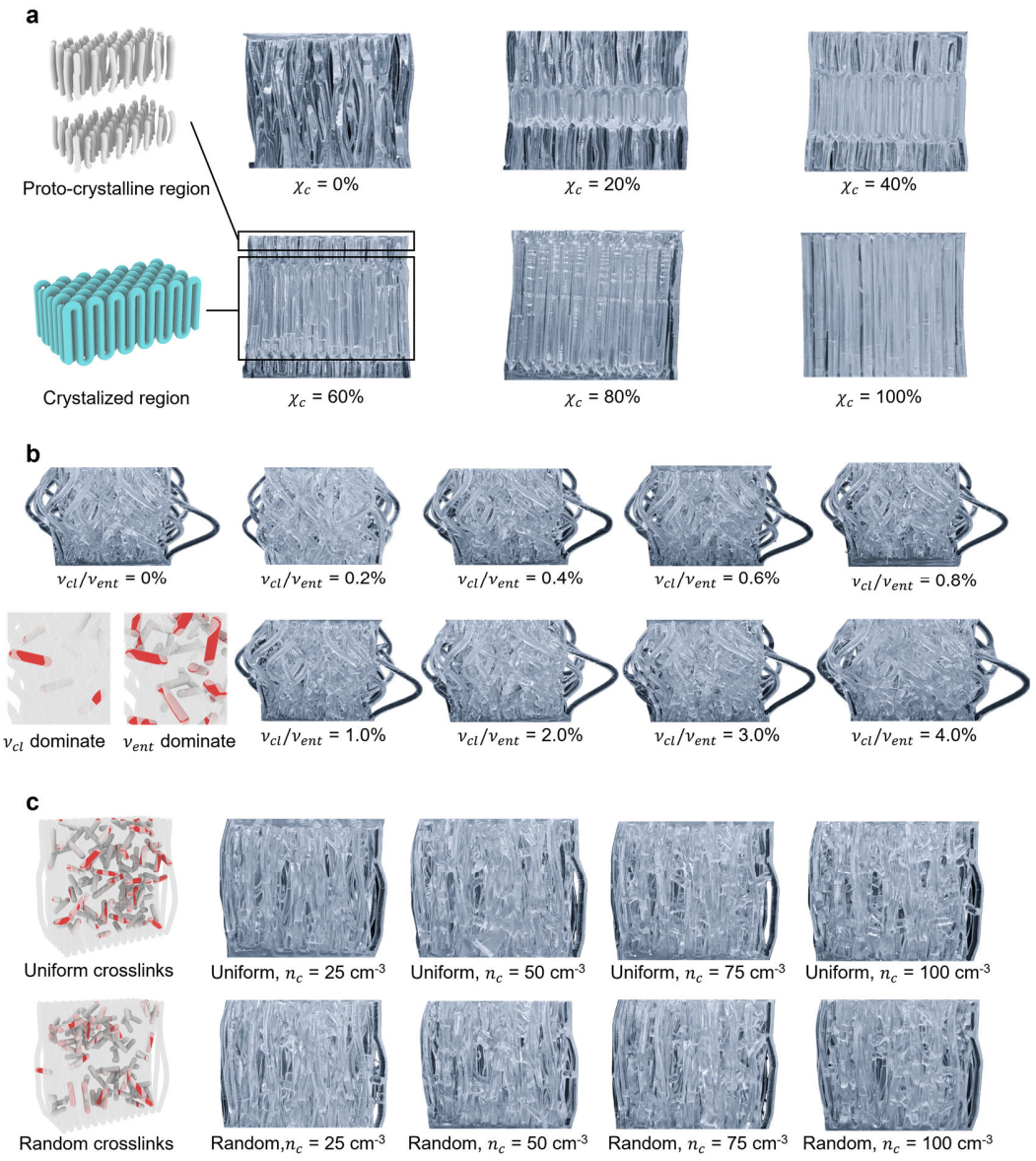
Supplementary Fig. 29. Chain energy partition under different entanglements per chain, ν_{ent}. Source data are provided as a Source Data file.



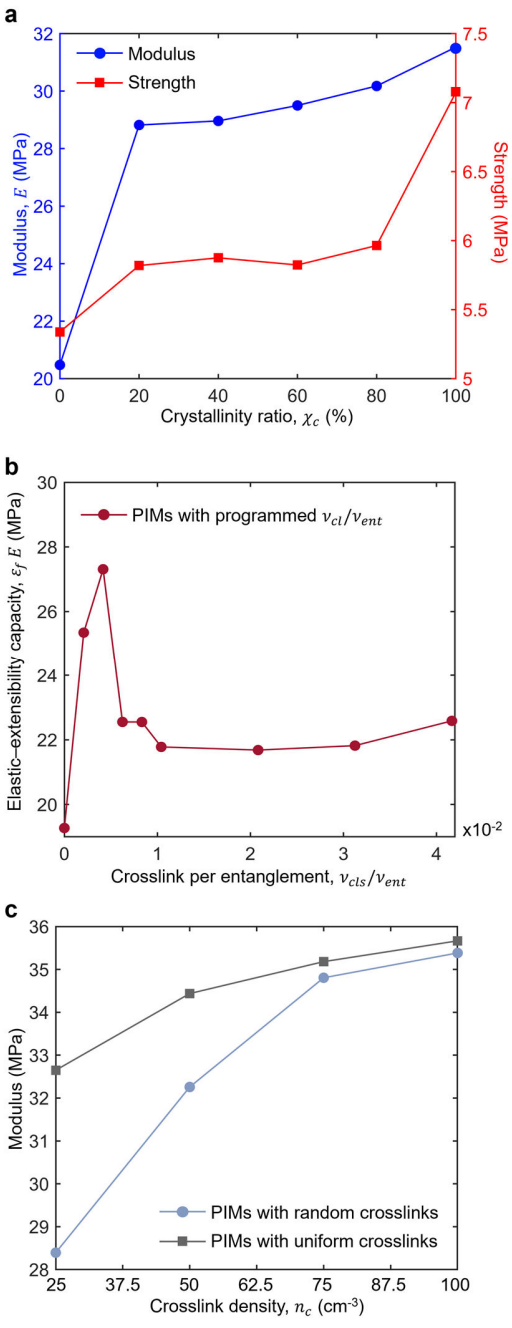
Supplementary Fig. 30. Chain energy partition under different proto-crystalline ordering, Ω. Source data are provided as a Source Data file.



Supplementary Fig. 31. Comparison between model predictions and experimental measurements on held-out PIM architectures using the energy-partition framework. The comparisons include (a) modulus, (b) strength, (c) fracture energy, and (d) fracture strain. Error bars denote standard deviations among three repetitive experiments. Source data are provided as a Source Data file.



Supplementary Fig. 32. PIMs with (a) different crystalline ratio (χ_c), (b) extensive entanglements per chain (ν_{ent}) with increasing crosslinks per chain (ν_{cl}), and (c) random and uniform crosslink distributions with increasing crosslink density n_c.



700

701 Supplementary Fig. 33. Experimental results of special PIM strengthening effects. This
702 includes experimentally measured (a) modulus and strength of crystalizing PIMs, (b)
703 elastic-extensibility capacity of PIMs losing entanglement domination over crosslinks,
704 where ν_{cl}, ν_{ent}, ε_f, and E are crosslinks per chain, entanglements per chain, fracture strain,
705 and modulus, respectively, and (c) modulus comparing PIMs with random and uniform
706 distribution of crosslinks at increasing crosslink density (n_c). Source data are provided as
707 a Source Data file.

Supplementary tables

Supplementary Table 1. The geometrical information of the fabricated samples.

Topology	Chain/bonding diameter (mm)	Density ($\times 10^{-3}$ g/cm^3)
PIM (MD strengthening)	1	122 – 199
PIM (PS strengthening)	1	160 – 165
PIM (CL strengthening)	1	164 – 431
CIM (BCC)	0.6 – 1.2	103 – 449
CIM (diamond)	0.6 – 1.2	110 – 517
CIM (FCC)	0.6 – 1.2	170 – 753
CIM (octet-truss)	0.6 – 1.2	254 – 1120
PIM+CIM	1	176 – 372
PIM enabled humanoid tissue	1	122 – 199

712 Supplementary Table 2. Summary of load scenarios and test regions for different types of
713 samples.

Type of sample	Cross-sectional area (mm^2)	Gauge length / compression size (mm)	Load type	Displacement rate (mm/min)	Mechanical properties
ASTM D638 type IV	18	20	Tension	2	Modulus, strength, fracture strain
PIM or CIM specimens	200	20	Tension / compression	1 / 1	Strain field, modulus, strength, energy absorption efficiency, fracture energy, fracture dissipation
PIM+CIM composite specimens	75 / 225	25 / 5	Tension / compression	1 / 1	Modulus, energy absorption capacity
Biometric exoskeleton	NA	105 / 20	Tension / Compression	1 / 1	Stimulated load
Multimodal robotic finger	NA	25 / 45	Tension / compressive-bending	1 / 1	Stimulated load
PIM enabled humanoid tissue	6.25	12	Tension	1	Strain field, average tissue strain, stimulated tissue stress, tissue modulus
Humanoid toe test specimen	NA	10	Compressive-bending / tensile-releasing	1	Reaction load, reaction slope
Humanoid finger test specimen	NA	10	Tensile-bending	1	Strain field, average tissue strain

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Supplementary References

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